# Cubical Holonomy, Explicit SU(2) Flux, and a Finite Gaussian Cusp Limit

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**Abstract**

We give a self-contained finite construction on a family of flat four-tori whose period matrices enter a controlled radial cusp. Two integers are kept independent: $M_{\mathrm{cov}}$, which changes the covering lattice, and $N_{\mathrm{mesh}}$, which refines a transported cubical decomposition. We identify the cubical vertices exactly with the finite torsion subgroup, type the continuum-to-link holonomy map and its gauge covariance, and exhibit a periodic diagonal SU(2) flux. Its logarithmic action is $2\pi^2 M_{\mathrm{cov}}^2/(g_{\mathrm{YM}}^2 D_T)$, while an explicitly defined continuum-matched $(1,2)$-sector Wilson functional has the same mesh limit. On the native fibre, a distinct smooth integral-flux SU(2) connection has the same numerical action but an exact background-gauge Jacobi eigenvalue $-4\pi/D_T$, so it is a classical saddle rather than a vacuum; no equality with the displayed link holonomies is asserted. The scalar Laplacian on every cover has an exact character spectrum; its deck-invariant sector is identified explicitly, and the cover momentum lattices become dense when both cusp and cover parameters diverge. A finite Coulomb-reduced cochain complex yields an exact Fourier spectrum and a positive Gaussian oscillator gap at every fixed triple $(T, M_{\mathrm{cov}}, N_{\mathrm{mesh}})$. Along $T_j = j^2$, $M_{\mathrm{cov},j} = j$, and $N_{\mathrm{mesh},j} = j^3$, the cover tori converge pointed locally isometrically to $\mathbb{R}^4$, the mesh resolves the cover geometry, and both the Gaussian gap and the flux action tend to zero. The direct integral of the mean-zero native scalar Laplacians over the radial parameter is a specified self-adjoint operator with trivial kernel and zero in its essential spectrum. These statements concern a finite quadratic oscillator and one explicitly normalized compact-link sector. On the native torus $M_{\mathrm{cov}} = 1$, we also identify the exact reduced continuum Yang–Mills Hessian gap and the exact finite gap in every mesh regime. Finally, explicit gauge-invariant probability measures show that the link kinematics and the displayed magnetic configuration determine no Wilson-loop area law, while bounded self-adjoint countermodels show that entanglement and a gapped Gaussian subspace determine no full spectral gap. We state and prove the full-form-domain coercive comparison that would transfer an auxiliary gap to a physical Hamiltonian. No such comparison is constructed here, so these results do not identify the oscillator with a nonlinear, gauge-invariant compact-link Yang–Mills Hamiltonian and do not constitute a physical mass-gap claim.

**Status and scope.** This is a standalone mathematical draft of exact finite geometric, compact-link, and Gaussian statements. All period hypotheses and normalizations used below are stated explicitly. No continuum quantum field theory is constructed; no transfer matrix, nonlinear Gauss-law Hamiltonian, renormalized limit, or Osterwalder–Schrader reconstruction is asserted. Consequently the Gaussian spectral gap proved here is not the spectral gap of physical four-dimensional compact-link Yang–Mills theory.

# 1 Introduction and exact standing assumptions

The purpose of this note is to separate, and then connect by explicit maps, three finite constructions associated with a degenerating family of flat four-tori. The first is geometric: a cover is equipped with the cubical decomposition transported from a coordinate torus. The second is genuinely compact-link: continuum connections produce group-valued edge holonomies, and an explicit periodic SU(2) flux has a computable plaquette action. The third is linear and Gaussian: a finite cochain complex on the same mesh produces a positive Coulomb-restricted Hessian and a finite harmonic oscillator.

The separation of parameters is essential. The cover index $M_{\mathrm{cov}}$ multiplies two period vectors and changes the global flat torus. The mesh integer $N_{\mathrm{mesh}}$ subdivides a fixed coordinate torus and changes only its cubical resolution. They are never identified. Only in the final diagonal sequence are both chosen as functions of a third integer $j$.

The period matrix and radial cusp specialization used below are taken from [1]. We restate every input needed here so that the proofs depend only on the explicit regular-fibre matrix and the quantified radial bounds below; neither the claimed compactification nor the claimed identification of its total space with $S^6$ is used. Every cover below is a fibrewise cover of the displayed regular flat torus. No finite cover of a compactified total space is asserted here.

Fix $T_0 > 0$. For every $T \geq T_0$, let $\tau_T, \mu_T, \beta_T \in \mathbb{C}$, and put

$$L_T = \operatorname{Im} \tau_T, \qquad m_T = \operatorname{Im} \mu_T, \qquad q_T = -\operatorname{Im} \beta_T, \qquad D_T = L_T q_T + 6m_T^2.$$

The standing period assumptions are the following exact quantified requirements: $D_T > 0$ for every $T \geq T_0$, and there is a constant $C_{\mathrm{per}} < \infty$, independent of $T$, such that

$$\begin{aligned} |L_T - T| \leq C_{\mathrm{per}}, \quad |q_T - T| \leq C_{\mathrm{per}}, \qquad |m_T| \leq C_{\mathrm{per}}, \\ |\operatorname{Re} \tau_T| \leq C_{\mathrm{per}}, \quad |\operatorname{Re} \mu_T| \leq C_{\mathrm{per}}, \quad |\operatorname{Re} \beta_T| \leq C_{\mathrm{per}}. \end{aligned} \tag{1.1}$$

Thus every $O(1)$ below is uniform for $T \geq T_0$, with a bound depending only on the fixed period family. We use the Euclidean metric on $\mathbb{R}^4$ and the induced quotient metrics. The gauge coupling $g_{\mathrm{YM}} > 0$ is fixed.

The ordered plaquette and real-trace structure used below is the classical Wilson construction [6]. Transfer-matrix and reflection positivity results such as [2, 3] require additional time/reflection structure, while Euclidean reconstruction requires the axiomatic input of [4, 5]. These references motivate the sharp boundary drawn in the last section; none is used to replace a proof of the finite claims.

### Results at a glance

Starting from the exact period matrix in (2.1), we prove:

1. the scalar Laplacian on the cover has a complete character spectrum, its deck-invariant Fourier sector is exact, and its momentum lattice has an explicit covering-radius bound;
2. the transported cubulation covers the entire cover torus, and its vertices are exactly, but only, the $N_{\mathrm{mesh}}$-torsion points;
3. continuum holonomy defines a typed gauge-equivariant map to link variables, with the endpoint inverses fixed by the stated ODE convention;

4. an explicit periodic diagonal SU(2) flux has constant $(1,2)$-plaquette and exact logarithmic and Wilson-sector actions;
5. on the native fibre, a separately constructed smooth integral-flux connection has the same numerical action and a completely typed background-gauge Jacobi operator with an explicit eigenvalue $-4\pi/D_T$; it is therefore a classical saddle, not a gapped vacuum;
6. the Coulomb-reduced finite Hessian has an exact Fourier spectrum, so its Gaussian quantization has a unique ground state and explicit gap;
7. on the native torus, the reduced continuum Hessian and the finite Hessian have exact first eigenvalues and multiplicities for every mesh size;
8. the direct integral of the native mean-zero scalar Laplacians over the radial parameter has trivial kernel but zero in its essential spectrum;
9. one diagonal limit simultaneously decompactifies the cover, resolves its mesh, and drives the finite Gaussian gap and flux action to zero; and
10. the exponential map is only a local chart, with an exact $N_{\mathrm{mesh}}^{-2}$ Hessian scaling, and supplies no global unitary identification with compact-link gauge theory;
11. gauge-invariant pure-gauge and product-Haar states on the same link configuration space give incompatible Wilson-loop expectations, so neither the kinematics nor the displayed flux selects a confinement law; and
12. entanglement is not a spectral sector, entangled eigenvalues can accumulate at zero, and an embedded gapped Gaussian range can coexist with a gapless orthogonal channel; a physical gap transfer requires a comparison on the entire physical excitation form domain.

## 2 Cubical holonomy and finite gauge systems

This section records three constructions on the regular cusp fibres while keeping two independent integers separate throughout: $M_{\mathrm{cov}}$ is the index used to enlarge two periods of a fibre, whereas $N_{\mathrm{mesh}}$ is the number of cubical subdivisions in each coordinate direction. In particular, no mesh refinement is identified with a change of covering space.

Let

$$P_T = \begin{pmatrix} 6\operatorname{Re}\mu_T & \operatorname{Re}\tau_T & 1 & 0 \\ 6\operatorname{Im}\mu_T & \operatorname{Im}\tau_T & 0 & 0 \\ \operatorname{Re}\beta_T & \operatorname{Re}\mu_T & 0 & 1 \\ \operatorname{Im}\beta_T & \operatorname{Im}\mu_T & 0 & 0 \end{pmatrix}, \qquad F_T = \mathbb{R}^4/P_T\mathbb{Z}^4. \tag{2.1}$$

Write

$$L_T = \operatorname{Im}\tau_T, \qquad m_T = \operatorname{Im}\mu_T, \qquad q_T = -\operatorname{Im}\beta_T, \qquad D_T = L_T q_T + 6m_T^2 > 0. \tag{2.2}$$

Expanding the determinant along the last two columns gives

$$\det P_T = L_T \operatorname{Im}\beta_T - 6m_T^2 = -D_T.$$

Along the positive radial cusp the source data give

$$L_T = T + O(1), \qquad q_T = T + O(1), \qquad m_T = O(1), \tag{2.3}$$

and all three real parts occurring in (2.1) are bounded. Equivalently, after ordering physical coordinates as $(x_1, x_3; x_2, x_4)$, define the upper block

$$A_T = \begin{pmatrix} 6\,\mathrm{Re}\,\mu_T & \mathrm{Re}\,\tau_T \\ \mathrm{Re}\,\beta_T & \mathrm{Re}\,\mu_T \end{pmatrix}.$$

The first two columns then have block form

$$\begin{pmatrix} A_T \\ B_T \end{pmatrix}, \qquad B_T = \begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix} = T \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + C_T, \qquad \sup_T \|C_T\|_{\mathrm{op}} < \infty. \tag{2.4}$$

For $M_{\mathrm{cov}} \geq 1$, put

$$S_{M_{\mathrm{cov}}} = \mathrm{diag}(1, 1, M_{\mathrm{cov}}, M_{\mathrm{cov}}), \qquad P_{T,M_{\mathrm{cov}}} = P_T S_{M_{\mathrm{cov}}}, \tag{2.5}$$

and define the degree-$M_{\mathrm{cov}}^2$ cover

$$\widetilde{F}_{T,M_{\mathrm{cov}}} = \mathbb{R}^4 / P_{T,M_{\mathrm{cov}}} \mathbb{Z}^4 \longrightarrow F_T. \tag{2.6}$$

Indeed, $S_{M_{\mathrm{cov}}} \mathbb{Z}^4 \subset \mathbb{Z}^4$ and

$$\mathbb{Z}^4 / S_{M_{\mathrm{cov}}} \mathbb{Z}^4 \cong (\mathbb{Z}/M_{\mathrm{cov}}\mathbb{Z})^2,$$

so the identity on $\mathbb{R}^4$ induces the asserted cover and its deck group has order $M_{\mathrm{cov}}^2$. Its Euclidean quotient metric, written on the coordinate torus $\mathbb{R}^4/\mathbb{Z}^4$, has Gram matrix and covolume

$$G_{T,M_{\mathrm{cov}}} = P_{T,M_{\mathrm{cov}}}^{\mathsf{T}} P_{T,M_{\mathrm{cov}}}, \qquad \mathcal{D}_{T,M_{\mathrm{cov}}} = \sqrt{\det G_{T,M_{\mathrm{cov}}}} = M_{\mathrm{cov}}^2 D_T. \tag{2.7}$$

## 2.1 Scalar characters and exact cover momentum sectors

Equip $\widetilde{F}_{T,M_{\mathrm{cov}}}$ with normalized Haar measure

$$\mathrm{d}\nu_{T,M_{\mathrm{cov}}} = \mathcal{D}_{T,M_{\mathrm{cov}}}^{-1}\, \mathrm{d\,vol}_{\widetilde{F}_{T,M_{\mathrm{cov}}}}. \tag{2.8}$$

Let $\widetilde{\Delta}_{T,M_{\mathrm{cov}}}$ be the nonnegative self-adjoint scalar Laplacian obtained from the closed form

$$\widetilde{\mathfrak{q}}_{T,M_{\mathrm{cov}}}[f] = \int_{\widetilde{F}_{T,M_{\mathrm{cov}}}} |\mathrm{d}f|^2\, \mathrm{d}\nu_{T,M_{\mathrm{cov}}}, \qquad \mathrm{Dom}(\widetilde{\mathfrak{q}}_{T,M_{\mathrm{cov}}}) = H^1, \tag{2.9}$$

whose operator domain is $H^2(\widetilde{F}_{T,M_{\mathrm{cov}}})$.

**Theorem 2.1** (Cover spectrum, invariant sector, and momentum density)**.** *The functions*

$$\widetilde{\chi}_{T,M_{\mathrm{cov}},k}([x]) = \exp\Big(2\pi i \left\langle P_T^{-\mathsf{T}} S_{M_{\mathrm{cov}}}^{-1} k, x \right\rangle\Big), \qquad k \in \mathbb{Z}^4, \tag{2.10}$$

*form a complete orthonormal basis, and*

$$\widetilde{\Delta}_{T,M_{\mathrm{cov}}} \widetilde{\chi}_{T,M_{\mathrm{cov}},k} = 4\pi^2 \left| P_T^{-\mathsf{T}} S_{M_{\mathrm{cov}}}^{-1} k \right|^2 \widetilde{\chi}_{T,M_{\mathrm{cov}},k}. \tag{2.11}$$

*If $c_{T,M_{\mathrm{cov}}} : \widetilde{F}_{T,M_{\mathrm{cov}}} \to F_T$ is the covering map and $\chi_{T,\ell}([x]) = \exp(2\pi i \langle P_T^{-\mathsf{T}} \ell, x \rangle)$, then*

$$c_{T,M_{\mathrm{cov}}}^{*} \chi_{T,\ell} = \widetilde{\chi}_{T,M_{\mathrm{cov}},S_{M_{\mathrm{cov}}}\ell}. \tag{2.12}$$

*Consequently, the deck-invariant sector is exactly*

$$k \in S_{M_{\mathrm{cov}}}\mathbb{Z}^4 \quad \Longleftrightarrow \quad k_3 \equiv k_4 \equiv 0 \pmod{M_{\mathrm{cov}}}. \tag{2.13}$$

*There is a constant $C_{\mathrm{mom}}$, independent of all sufficiently large $T$ and all $M_{\mathrm{cov}} \geq 1$, such that*

$$\sup_{p\in\mathbb{R}^4} \operatorname{dist}\Big(p, P_T^{-\mathsf{T}} S_{M_{\mathrm{cov}}}^{-1}\mathbb{Z}^4\Big) \leq C_{\mathrm{mom}}\left(\frac{1}{T} + \frac{1}{M_{\mathrm{cov}}}\right). \tag{2.14}$$

*Thus, whenever $T_j, M_{\mathrm{cov},j} \to \infty$, these momentum lattices become locally dense in $\mathbb{R}^4$, and every $E \geq 0$ is a limit of eigenvalues in (2.11).*

*Proof.* The dual lattice of $P_T S_{M_{\mathrm{cov}}}\mathbb{Z}^4$ is

$$(P_T S_{M_{\mathrm{cov}}})^{-\mathsf{T}}\mathbb{Z}^4 = P_T^{-\mathsf{T}} S_{M_{\mathrm{cov}}}^{-1}\mathbb{Z}^4. \tag{2.15}$$

A character of $\mathbb{R}^4$ descends to the quotient precisely when its frequency lies in this dual lattice. Fourier theory for the compact abelian group, normalized by (2.8), gives the complete orthonormal basis (2.10); applying the positive Euclidean Laplacian proves (2.11). Substituting $k = S_{M_{\mathrm{cov}}}\ell$ proves (2.12). A cover character descends exactly when it is deck invariant, so (2.12) and Fourier completeness prove (2.13).

For the quantitative assertion, abbreviate

$$\alpha_T = \operatorname{Re}\mu_T, \qquad \eta_T = \operatorname{Re}\tau_T, \qquad c_T = \operatorname{Re}\beta_T, \qquad M = M_{\mathrm{cov}}.$$

Solving $P_T^{\mathsf{T}} x = e_j$ gives the four columns

$$\begin{aligned}
b_1 &= \left(0, \frac{m_T}{D_T}, 0, -\frac{L_T}{D_T}\right),\\
b_2 &= \left(0, \frac{q_T}{D_T}, 0, \frac{6m_T}{D_T}\right),\\
b_3 &= \frac{1}{M}\left(1, \frac{-6m_T\alpha_T - q_T\eta_T}{D_T}, 0, \frac{6(L_T\alpha_T - m_T\eta_T)}{D_T}\right),\\
b_4 &= \frac{1}{M}\left(0, \frac{-m_T c_T - q_T\alpha_T}{D_T}, 1, \frac{L_T c_T - 6m_T\alpha_T}{D_T}\right)
\end{aligned}$$

of $P_T^{-\mathsf{T}} S_M^{-1}$. The standing bounds imply $D_T \asymp T^2$ and hence, for a uniform $C_0$,

$$|b_1| + |b_2| \leq \frac{C_0}{T}, \qquad |b_3| + |b_4| \leq \frac{C_0}{M}.$$

Write any $p \in \mathbb{R}^4$ uniquely as $p = \sum_j t_j b_j$, and choose $n_j \in \mathbb{Z}$ with $|t_j - n_j| \leq \frac{1}{2}$. Then

$$\left|p - \sum_j n_j b_j\right| \leq \frac{1}{2}\sum_j |b_j| \leq \frac{C_0}{2}\left(\frac{1}{T} + \frac{1}{M}\right),$$

which proves (2.14). If $E > 0$, choose $p$ with $4\pi^2|p|^2 = E$ and approximate it by the cover momentum lattice; continuity gives eigenvalues tending to $E$. For $E = 0$, the index $k = (1,0,0,0)^{\mathsf{T}}$ has momentum norm $O(T^{-1})$. □

**Remark 2.2** (Fibrewise scope)**.** Theorem 2.1 is entirely fibrewise. It does not assert that these covers assemble compatibly with monodromy over a punctured base or extend across a singular fibre.

**Proposition 2.3** (Native and expanding-cover pointed limits)**.** *Let*

$$\mathbb{T}^2_{\mathrm{re}} = (\mathbb{R}e_1 \oplus \mathbb{R}e_3)/(\mathbb{Z}e_1 \oplus \mathbb{Z}e_3).$$

*Then the pointed native tori satisfy*

$$(F_T, 0) \longrightarrow (\mathbb{T}^2_{\mathrm{re}} \times (\mathbb{R}e_2 \oplus \mathbb{R}e_4), (0,0)) \cong (\mathbb{T}^2 \times \mathbb{R}^2, (0,0)), \tag{2.16}$$

*in pointed smooth topology: each fixed-radius ball is eventually exactly isometric to the corresponding ball in the limit. There are constants $C, C' > 0$ such that, for all sufficiently large $T$ and every $M \geq 1$,*

$$\min\{T - C, M\} \leq \mathrm{sys}(\widetilde{F}_{T,M}) \leq \min\{T + C', M\}. \tag{2.17}$$

*Consequently, for every integer-valued $M(T) \to \infty$,*

$$(\widetilde{F}_{T,M(T)}, 0) \longrightarrow (\mathbb{R}^4, 0) \tag{2.18}$$

*with the same exact fixed-ball property. In particular, the native and expanding-cover unscaled pointed limits are different.*

*Proof.* The period lattice of $F_T$ is generated by the first two columns of $P_T$, denoted $v_{\gamma,T}, v_{u,T}$, together with $e_1, e_3$. First quotient by $\mathbb{Z}e_1 \oplus \mathbb{Z}e_3$. The projection of $av_{\gamma,T} + bv_{u,T}$ to the noncompact $(e_2, e_4)$-plane is

$$B_T \begin{pmatrix} a \\ b \end{pmatrix} = \left( T \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + C_T \right) \begin{pmatrix} a \\ b \end{pmatrix}.$$

With $C = \sup_T \|C_T\|_{\mathrm{op}}$, its norm is at least $(T - C)\sqrt{a^2 + b^2}$. Hence the remaining nontrivial quotient translations eventually displace every point by more than $4R$. The quotient map is then injective and distance-preserving on the radius-$R$ ball, proving (2.16).

Every nonzero vector in the cover lattice has the form

$$av_{\gamma,T} + bv_{u,T} + M(pe_1 + qe_3), \qquad a, b, p, q \in \mathbb{Z}.$$

If $(a, b) \neq (0, 0)$, its noncompact-plane projection has norm at least $T - C$; if $(a, b) = (0, 0)$, its norm is at least $M$. This gives the lower bound in (2.17). The vector $Me_1$ has length $M$, while either long generator has length at most $T + C'$ for a uniform $C'$, proving the upper bound. If both $T$ and $M(T)$ diverge, the systole diverges. Once it exceeds $4R$, the universal covering map identifies the radius-$R$ balls isometrically, proving (2.18). □

## 2.2 The transported cubical subdivision and continuum holonomy

Fix an integer $N_{\mathrm{mesh}} \geq 2$, and let $K_{N_{\mathrm{mesh}}}$ be the standard cubical subdivision of $\mathbb{R}^4/\mathbb{Z}^4$ obtained by dividing every coordinate circle into $N_{\mathrm{mesh}}$ equal intervals. Its vertex set and positively oriented coordinate edges are

$$V_{N_{\mathrm{mesh}}} = (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \qquad E_{N_{\mathrm{mesh}}} = V_{N_{\mathrm{mesh}}} \times \{1, 2, 3, 4\}. \tag{2.19}$$

Here and below, $\mathbf{e}_1, \ldots, \mathbf{e}_4$ denotes the standard basis of $\mathbb{R}^4$, reduced modulo $N_{\mathrm{mesh}}$ when used to translate an element of $V_{N_{\mathrm{mesh}}}$.

**Proposition 2.4** (Exact cubical realization and the torsion vertices)**.** *The linear map $P_{T,M_{\mathrm{cov}}}$ induces a diffeomorphism*

$$\Phi_{T,M_{\mathrm{cov}}} : \mathbb{R}^4/\mathbb{Z}^4 \longrightarrow \widetilde{F}_{T,M_{\mathrm{cov}}}, \qquad [y] \longmapsto [P_{T,M_{\mathrm{cov}}} y]. \tag{2.20}$$

*Transporting $K_{N_{\mathrm{mesh}}}$ by this diffeomorphism gives an honest cubical cell decomposition of the whole torus. On vertices its restriction is the bijection*

$$\begin{aligned} \iota_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} : \quad & (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4 \xrightarrow{\cong} \widetilde{F}_{T,M_{\mathrm{cov}}}[N_{\mathrm{mesh}}], \\ & [n] \longmapsto \left[\frac{1}{N_{\mathrm{mesh}}} P_{T,M_{\mathrm{cov}}} n\right]. \end{aligned} \tag{2.21}$$

*Here the codomain is the finite $N_{\mathrm{mesh}}$-torsion subgroup, not the entire torus. The coordinate edge $(n, j)$ is the path*

$$\gamma_{n,j}(t) = \left[\frac{1}{N_{\mathrm{mesh}}} P_{T,M_{\mathrm{cov}}}(n + t\mathbf{e}_j)\right], \qquad 0 \le t \le 1. \tag{2.22}$$

*More generally, for an ordered subset $I = \{i_1, \dots, i_p\} \subset \{1, 2, 3, 4\}$, the corresponding closed $p$-cell is parametrized by*

$$c_{n,I}(t_1, \dots, t_p) = \left[\frac{1}{N_{\mathrm{mesh}}} P_{T,M_{\mathrm{cov}}} \left(n + \sum_{a=1}^{p} t_a \mathbf{e}_{i_a}\right)\right], \qquad (t_1, \dots, t_p) \in [0, 1]^p. \tag{2.23}$$

*Proof.* If $y - y' \in \mathbb{Z}^4$, then $P_{T,M_{\mathrm{cov}}}(y - y')$ belongs to the quotient lattice, so (2.20) is well defined. The matrix $P_{T,M_{\mathrm{cov}}}$ is invertible, and its inverse induces the inverse map on the two quotients. Hence it is a diffeomorphism and carries every open cell of the standard subdivision diffeomorphically onto an open cell of the target.

Every point displayed in (2.21) is killed by $N_{\mathrm{mesh}}$. If it is zero, then

$$\frac{P_{T,M_{\mathrm{cov}}} n}{N_{\mathrm{mesh}}} \in P_{T,M_{\mathrm{cov}}} \mathbb{Z}^4.$$

Invertibility gives $n \in N_{\mathrm{mesh}}\mathbb{Z}^4$, proving injectivity. Conversely, if $[x]$ is killed by $N_{\mathrm{mesh}}$, choose a lift $x \in \mathbb{R}^4$. There is $n \in \mathbb{Z}^4$ such that $N_{\mathrm{mesh}} x = P_{T,M_{\mathrm{cov}}} n$, and therefore $[x] = [P_{T,M_{\mathrm{cov}}} n / N_{\mathrm{mesh}}]$. This proves surjectivity. Formula (2.22) is simply the transport of the corresponding coordinate edge; at a wrap-around endpoint the extra vector $P_{T,M_{\mathrm{cov}}} \mathbf{e}_j$ is a period. □

Let $G$ be a compact matrix Lie group and use the trivial principal bundle over $\widetilde{F}_{T,M_{\mathrm{cov}}}$. A connection is represented by $A \in \Omega^1(\widetilde{F}_{T,M_{\mathrm{cov}}}; \mathfrak{g})$. For an oriented path $\gamma$, define $U_A(\gamma)$ by

$$\frac{dU}{dt} = -A_{\gamma(t)}(\dot{\gamma}(t)) U(t), \qquad U(0) = 1, \qquad U_A(\gamma) = U(1). \tag{2.24}$$

**Proposition 2.5** (Typed continuum-to-link holonomy map)**.** *The formula*

$$\mathrm{Hol}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(A)_{n,j} = U_A(\gamma_{n,j}) \tag{2.25}$$

*defines a map*

$$\mathrm{Hol}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} : \Omega^1(\widetilde{F}_{T,M_{\mathrm{cov}}}; \mathfrak{g}) \longrightarrow G^{E_{N_{\mathrm{mesh}}}}. \tag{2.26}$$

*For the convention*

$$A^g = g^{-1} A g + g^{-1} dg, \qquad g \in C^\infty(\widetilde{F}_{T,M_{\mathrm{cov}}}, G), \tag{2.27}$$

*it is equivariant for evaluation of $g$ at the vertices:*

$$\mathrm{Hol}(A^g)_{n,j} = g(\gamma_{n,j}(1))^{-1} \mathrm{Hol}(A)_{n,j} g(\gamma_{n,j}(0)). \tag{2.28}$$

*Proof.* The smooth matrix ODE (2.24) has a unique solution on every edge, which proves that (2.26) is defined. Let $U$ solve that ODE for $A$, set $g_t = g(\gamma_{n,j}(t))$, and put $V(t) = g_t^{-1}U(t)g_0$. Direct differentiation gives

$$V'(t) = -(g_t^{-1}A(\dot\gamma_{n,j})g_t + g_t^{-1}g_t')V(t), \qquad V(0) = 1.$$

This is exactly the parallel-transport equation for $A^g$. Uniqueness and evaluation at $t = 1$ prove (2.28). Notice that the endpoint inverse in this formula is forced by the joint conventions (2.24) and (2.27). □

### 2.3 An explicit compact-link SU(2) flux

The ordered plaquette convention used below is most naturally written for the inverse-transport link variable

$$\mathcal{U}(A)_{n,j} = \mathrm{Hol}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(A)_{n,j}^{-1}. \tag{2.29}$$

Equation (2.28) then becomes

$$\mathcal{U}(A^g)_{n,j} = g(\gamma_{n,j}(0))^{-1}\mathcal{U}(A)_{n,j}g(\gamma_{n,j}(1)). \tag{2.30}$$

Thus the product in (2.36) transforms by conjugation at its base vertex. The symbols $U_j(n)$ in the remainder of this subsection use this inverse-transport convention.

Use

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) : X^* = -X,\ \mathrm{tr}\, X = 0\}, \qquad \langle X, Y\rangle_{\mathfrak{su}(2)} = -\frac{1}{2}\,\mathrm{tr}(XY), \tag{2.31}$$

on real elements, and fix

$$H_{\mathrm{link}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \|H_{\mathrm{link}}\|^2 = 1, \qquad \exp(2\pi H_{\mathrm{link}}) = 1. \tag{2.32}$$

Write $N = N_{\mathrm{mesh}}$ temporarily inside the next displayed link formulas, and represent $n_1, n_2$ by integers in $\{0, \dots, N-1\}$.

The link field constructed next is an abstract compact-link configuration. It is not asserted to lie in the image of the smooth-connection holonomy map (2.26).

**Theorem 2.6** (A periodic compact-link flux and its exact action)**.** *On the oriented edges of $K_N$, define*

$$U_1(n) = \exp\left(-\frac{2\pi n_2}{N^2}H_{\mathrm{link}}\right), \tag{2.33}$$

$$U_2(n) = \begin{cases} 1, & 0 \le n_2 \le N-2, \\ \exp(2\pi n_1 H_{\mathrm{link}}/N), & n_2 = N-1, \end{cases} \tag{2.34}$$

$$U_3(n) = U_4(n) = 1. \tag{2.35}$$

*For*

$$U_{ij}(n) = U_i(n)U_j(n+\mathbf{e}_i)U_i(n+\mathbf{e}_j)^{-1}U_j(n)^{-1}, \tag{2.36}$$

*one has*

$$\begin{aligned} &U_{12}(n) = \exp(2\pi H_{\mathrm{link}}/N^2) && \text{for every } n, \\ &U_{ij}(n) = 1 && \text{for } 1 \le i < j \le 4, \quad (i,j) \ne (1,2). \end{aligned} \tag{2.37}$$

*Let* Log *be the principal logarithm on the displayed plaquette. It is unambiguous because* $0 < 2\pi/N^2 \le \pi/2$. *Define the logarithmic curvature two-form and its metric-weighted action by*

$$\mathcal{F}_N^{\log}(n) = N^2 \operatorname{Log}(U_{12}(n))\, dy^1 \wedge dy^2, \tag{2.38}$$

$$S^{\log}_{T,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}}{2g_{\mathrm{YM}}^2 N^4} \sum_{n\in V_N} \|\mathcal{F}_N^{\log}(n)\|^2_{G^{-1}_{T,M_{\mathrm{cov}}}\otimes\mathfrak{su}(2)}. \tag{2.39}$$

*Then*

$$\mathcal{F}_N^{\log}(n) = 2\pi H_{\mathrm{link}}\, dy^1 \wedge dy^2, \qquad S^{\log}_{T,M_{\mathrm{cov}},N}(U) = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}. \tag{2.40}$$

*In particular, the native-fibre value* $M_{\mathrm{cov}} = 1$ *is* $2\pi^2/(g_{\mathrm{YM}}^2 D_T)$, *independently of the mesh.*

*There is no canonical scalar Wilson action for a general nonorthogonal metric obtained merely by assigning independent character weights to all coordinate plaquettes. For this configuration, however, the exact continuum-matched* $(1,2)$*-sector Wilson functional is unambiguous if it is defined explicitly by the following formula. Its ordered plaquette holonomy and real trace are the Wilson lattice-action construction [6]; the metric coefficient and normalization here are the present definition, and their consequences are proved below rather than imported from that source:*

$$S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}\kappa^{12}_{T,M_{\mathrm{cov}}}}{2g_{\mathrm{YM}}^2} \sum_{n\in V_N} \left(2 - \operatorname{Re}\operatorname{tr} U_{12}(n)\right), \tag{2.41}$$

*where*

$$\kappa^{12}_{T,M_{\mathrm{cov}}} = (G^{-1}_{T,M_{\mathrm{cov}}})_{11}(G^{-1}_{T,M_{\mathrm{cov}}})_{22} - (G^{-1}_{T,M_{\mathrm{cov}}})^2_{12} = D_T^{-2}. \tag{2.42}$$

*Its exact value and continuum-mesh limit are*

$$S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{M_{\mathrm{cov}}^2 N^4}{2g_{\mathrm{YM}}^2 D_T}\left(2 - 2\cos\frac{2\pi}{N^2}\right), \tag{2.43}$$

$$= \frac{M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}\left(2\pi^2 - \frac{2\pi^4}{3N^4} + \frac{4\pi^6}{45N^8} + O(N^{-12})\right), \tag{2.44}$$

$$\lim_{N\to\infty} S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}. \tag{2.45}$$

*Proof.* All links lie in the same diagonal circle subgroup, so they commute. If $0 \le n_2 \le N-2$, the two $U_2$-terms in (2.36) are trivial and

$$U_{12}(n) = \exp(-2\pi n_2 H_{\mathrm{link}}/N^2)\exp(2\pi(n_2+1)H_{\mathrm{link}}/N^2) = \exp(2\pi H_{\mathrm{link}}/N^2).$$

If $n_2 = N-1$, then $U_1(n+\mathbf{e}_2) = 1$. Moreover,

$$U_2(n+\mathbf{e}_1)U_2(n)^{-1} = \exp(2\pi H_{\mathrm{link}}/N).$$

For $n_1 = N-1$, the exponent computed with representatives differs from $2\pi H_{\mathrm{link}}/N$ by $-2\pi H_{\mathrm{link}}$, whose exponential is one; hence the formula also holds across that wrap boundary. It follows that

$$U_{12}(n) = \exp\left(-\frac{2\pi(N-1)}{N^2}H_{\mathrm{link}}\right)\exp\left(\frac{2\pi}{N}H_{\mathrm{link}}\right) = \exp(2\pi H_{\mathrm{link}}/N^2).$$

The functions $U_1, U_2$ are independent of $n_3, n_4$, and $U_3 = U_4 = 1$, proving every remaining assertion in (2.37). Reversing the orientation gives $U_{21}(n) = U_{12}(n)^{-1}$; it is not an additional trivial plaquette.

The lower-right $2 \times 2$ principal block of the cover Gram matrix is $M_{\mathrm{cov}}^2 I_2$. Jacobi's complementary-minor identity therefore gives

$$\kappa^{12}_{T,M_{\mathrm{cov}}} = \frac{\det(M_{\mathrm{cov}}^2 I_2)}{\det G_{T,M_{\mathrm{cov}}}} = \frac{M_{\mathrm{cov}}^4}{M_{\mathrm{cov}}^4 D_T^2} = D_T^{-2},$$

which proves (2.42). By (2.32) and the principal-log choice, $N^2 \operatorname{Log} U_{12} = 2\pi H_{\mathrm{link}}$. Hence every one of the $N^4$ summands in (2.39) equals $4\pi^2 D_T^{-2}$. Substitution of $\mathcal{D}_{T,M_{\mathrm{cov}}} = M_{\mathrm{cov}}^2 D_T$ proves (2.40).

Finally, $H_{\mathrm{link}}$ has eigenvalues $\pm i$, so

$$\operatorname{Re} \operatorname{tr} \exp(2\pi H_{\mathrm{link}}/N^2) = 2\cos(2\pi/N^2).$$

There are $N^4$ equal plaquettes. Substitution proves (2.43). Applying

$$1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + \frac{x^6}{720} + O(x^8)$$

with $x = 2\pi/N^2$ proves (2.44) and (2.45). □

### 2.4 A distinct smooth native magnetic saddle

The compact-link field above was defined directly on edges. We now construct a smooth integral-flux connection on the native torus $F_T$. Its action equals the native value in (2.40), but equality of two real numbers is not an identification of connections: no claim is made that the links (2.33)–(2.35) are the holonomies of the connection below.

Use the physical coordinate order $x = (x_1, y_1, x_2, y_2)^{\mathsf{T}}$, put $u = P_T^{-1} x$, and define the integral coframe

$$\theta_T^j = du_j = (P_T^{-\mathsf{T}} \mathbf{e}_j)^{\mathsf{T}} dx. \tag{2.46}$$

For the marked cycle $s \mapsto [x + sP_T \mathbf{e}_k]$, one has $\int \theta_T^j = \delta_{jk}$. Direct solution of $P_T^{\mathsf{T}} \xi = \mathbf{e}_1, \mathbf{e}_2$, without replacing the skew lattice by a rectangle, gives

$$\theta_T^1 = \frac{m_T}{D_T}\, dy_1 - \frac{L_T}{D_T}\, dy_2, \qquad \theta_T^2 = \frac{q_T}{D_T}\, dy_1 + \frac{6m_T}{D_T}\, dy_2. \tag{2.47}$$

Consequently,

$$\theta_T^1 \wedge \theta_T^2 = \frac{1}{D_T}\, dy_1 \wedge dy_2, \qquad |\theta_T^1 \wedge \theta_T^2|^2 = D_T^{-2}. \tag{2.48}$$

Indeed, the wedge coefficient is $(6m_T^2 + L_T q_T)/D_T^2 = D_T^{-1}$, and $dy_1, dy_2$ are orthonormal for the fixed Euclidean quotient metric.

For $n \in \mathbb{Z}^4$, set

$$j_n(u) = \exp(-2\pi i n_1 u_2). \tag{2.49}$$

The identity

$$j_m(u+n) j_n(u) = j_{m+n}(u) e^{-2\pi i m_1 n_2} = j_{m+n}(u)$$

shows that

$$\mathscr{L}_T = (\mathbb{R}^4 \times \mathbb{C})/\mathbb{Z}^4, \qquad n \cdot (u, z) = (u + n, j_n(u) z), \tag{2.50}$$

is a Hermitian line bundle on $F_T$. On its pullback, the unitary connection

$$\widetilde{\nabla}^{\mathscr{L}_T} = d + 2\pi i u_1\, du_2 \tag{2.51}$$

descends. In fact, for $\mathcal{A} = 2\pi i u_1 du_2$,

$$\mathcal{A}(u+n) = \mathcal{A}(u) + 2\pi i n_1 du_2, \qquad j_n^{-1} dj_n = -2\pi i n_1 du_2,$$

which is exactly the descent identity. Its curvature is

$$F_{\mathscr{L}_T} = 2\pi i\, du_1 \wedge du_2 = 2\pi i\, \theta_T^1 \wedge \theta_T^2. \tag{2.52}$$

Let $\mathscr{E}_T = \mathscr{L}_T \oplus \mathscr{L}_T^{-1}$. Its transition matrices are

$$G_n(u) = \begin{pmatrix} j_n(u) & 0 \\ 0 & j_n(u)^{-1} \end{pmatrix} \in \mathrm{SU}(2),$$

so evaluation supplies a parallel determinant trivialization. Fix the same normalization as in (2.31), and write

$$H_{\mathrm{mag}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad |H_{\mathrm{mag}}|^2 = 1. \tag{2.53}$$

The direct-sum connection has the lifted expression

$$\nabla^T = d + 2\pi u_1\, du_2\, H_{\mathrm{mag}}, \qquad F_{\nabla^T} = 2\pi H_{\mathrm{mag}} \theta_T^1 \wedge \theta_T^2 = \frac{2\pi}{D_T} H_{\mathrm{mag}}\, dy_1 \wedge dy_2. \tag{2.54}$$

**Proposition 2.7** (Smooth critical connection and exact action)**.** *The connection $\nabla^T$ is non-flat and satisfies $d^*_{\nabla^T} F_{\nabla^T} = 0$. For*

$$\mathrm{YM}_T(\nabla) = \frac{1}{2g_{\mathrm{YM}}^2} \int_{F_T} |F_\nabla|^2\, d\,\mathrm{vol}_{F_T}, \tag{2.55}$$

*one has*

$$\mathrm{YM}_T(\nabla^T) = \frac{2\pi^2}{g_{\mathrm{YM}}^2 D_T} = \frac{2\pi^2}{g_{\mathrm{YM}}^2 T^2}(1 + O(T^{-1})). \tag{2.56}$$

*This is exactly the $M_{\mathrm{cov}} = 1$ value of (2.40).*

*Proof.* The transition matrices commute with $H_{\mathrm{mag}}$, so this diagonal endomorphism is a global parallel section of $\mathrm{ad}\,\mathscr{E}_T$. The coframe is parallel for the flat metric. Hence $\nabla^T F_{\nabla^T} = 0$, which implies the Yang–Mills equation. The curvature is nonzero because $D_T > 0$. By (2.48), (2.7) at $M_{\mathrm{cov}} = 1$, and $|H_{\mathrm{mag}}| = 1$,

$$\int_{F_T} |F_{\nabla^T}|^2\, d\,\mathrm{vol}_{F_T} = 4\pi^2 D_T D_T^{-2} = \frac{4\pi^2}{D_T}.$$

This proves the exact action. The asymptotic follows from $D_T = T^2 + O(T)$, which is a consequence of (1.1). □

We next specify the background-gauge Jacobi operator, including every closed domain used below. Let

$$\mathscr{H}_T^{\mathrm{ad},p} = L^2\Omega^p(F_T; \mathrm{ad}\,\mathscr{E}_T), \qquad p = 0, 1,$$

and complexify these real Hilbert spaces for the resolvent argument. All operators commute with the real structure. Define the maximal distributional codifferential by

$$\begin{aligned}\mathrm{Dom}(\delta_T^{\max}) = \{a \in \mathscr{H}_T^{\mathrm{ad},1} :\ &\text{there is } u \in \mathscr{H}_T^{\mathrm{ad},0} \text{ such that} && (2.57)\\ &\langle a, d_{\nabla^T}\phi\rangle_{L^2} = \langle u, \phi\rangle_{L^2} \text{ for all } \phi \in C^\infty(F_T; \mathrm{ad}\,\mathscr{E}_T)\}, && (2.58)\\ &\delta_T^{\max} a = u.\end{aligned}$$

Density of the test space makes $u$ unique. Passing to the limit in the testing identity proves that $\delta_T^{\max}$ is closed. Therefore

$$\mathscr{S}_T = \ker \delta_T^{\max} \tag{2.59}$$

is a closed background Coulomb slice.

The closed scalar form

$$q_T^{(0)}[\phi,\psi] = \langle d_{\nabla^T}\phi, d_{\nabla^T}\psi\rangle_{L^2}, \qquad \mathrm{Dom}(q_T^{(0)}) = H^1\Omega^0(F_T; \mathrm{ad}\,\mathscr{E}_T),$$

has the nonnegative Friedrichs operator

$$\Delta_{\nabla^T}^{(0)} = d_{\nabla^T}^* d_{\nabla^T}, \qquad \mathrm{Dom}(\Delta_{\nabla^T}^{(0)}) = H^2\Omega^0(F_T; \mathrm{ad}\,\mathscr{E}_T). \tag{2.60}$$

The $H^2$ equality is elliptic regularity on the compact flat torus.

In a constant oriented orthonormal frame $(e_\mu)_{\mu=1}^4$, write $F_{\nu\mu} = F_{\nabla^T}(e_\nu, e_\mu)$, and define

$$(V_T a)_\nu = -2\sum_{\mu=1}^4 [F_{\nu\mu}, a_\mu].$$

Antisymmetry of $F_{\mu\nu}$ and skew-adjointness of $\mathrm{ad}(F_{\nu\mu})$ show that $V_T$ is symmetric. It is a bounded bundle endomorphism. Hence

$$\begin{aligned}\mathfrak{j}_T^{\mathrm{gf}}[a,b] &= \langle \nabla^T a, \nabla^T b\rangle_{L^2} + \langle V_T a, b\rangle_{L^2}, && (2.61)\\ \mathrm{Dom}(\mathfrak{j}_T^{\mathrm{gf}}) &= H^1\Omega^1(F_T; \mathrm{ad}\,\mathscr{E}_T)\end{aligned}$$

is closed and lower bounded. Its self-adjoint operator $\mathcal{J}_T^{\mathrm{gf}}$ has

$$\mathrm{Dom}(\mathcal{J}_T^{\mathrm{gf}}) = H^2\Omega^1(F_T; \mathrm{ad}\,\mathscr{E}_T), \tag{2.62}$$

$$(\mathcal{J}_T^{\mathrm{gf}} a)_\nu = -\sum_{\mu=1}^4 \nabla_{e_\mu}\nabla_{e_\mu} a_\nu - 2\sum_{\mu=1}^4 [F_{\nu\mu}, a_\mu]. \tag{2.63}$$

**Proposition 2.8** (The Coulomb slice is reducing)**.** *For every $z \in \mathbb{C}\setminus\mathbb{R}$ and $f \in \mathrm{Dom}(\delta_T^{\max})$,*

$$\delta_T^{\max}(\mathcal{J}_T^{\mathrm{gf}} - z)^{-1} f = (\Delta_{\nabla^T}^{(0)} - z)^{-1}\delta_T^{\max} f. \tag{2.64}$$

*Thus $\mathscr{S}_T$ reduces $\mathcal{J}_T^{\mathrm{gf}}$, and its self-adjoint part $\mathcal{J}_T$ has the exact operator and form domains*

$$\begin{aligned}\mathcal{J}_T &= \mathcal{J}_T^{\mathrm{gf}}|_{\mathscr{S}_T},\\ \mathrm{Dom}(\mathcal{J}_T) &= H^2\Omega^1(F_T; \mathrm{ad}\,\mathscr{E}_T) \cap \mathscr{S}_T, && (2.65)\\ \mathrm{Dom}(\mathfrak{j}_T) &= H^1\Omega^1(F_T; \mathrm{ad}\,\mathscr{E}_T) \cap \mathscr{S}_T.\end{aligned}$$

*Proof.* Write $\nabla_\mu = \nabla^T_{e_\mu}$. For a smooth adjoint-valued scalar $\phi$, the commutator $[\nabla_\mu, \nabla_\nu]\phi = [F_{\mu\nu}, \phi]$, with both curvature terms retained, gives

$$\begin{aligned}&\big(\mathcal{J}_T^{\mathrm{gf}} d_{\nabla^T}\phi - d_{\nabla^T}\Delta^{(0)}_{\nabla^T}\phi\big)_\nu \\ &\quad = -\nabla_\mu\nabla_\mu\nabla_\nu\phi + \nabla_\nu\nabla_\mu\nabla_\mu\phi - 2[F_{\nu\mu}, \nabla_\mu\phi] \\ &\quad = -[\nabla_\mu F_{\mu\nu}, \phi] = 0,\end{aligned}$$

where the last equality is the Yang–Mills equation from Proposition 2.7. Formal adjunction gives

$$d^*_{\nabla^T}\mathcal{J}_T^{\mathrm{gf}} = \Delta^{(0)}_{\nabla^T} d^*_{\nabla^T}$$

on smooth one-forms.

Fix $z \notin \mathbb{R}$, let $f \in \mathrm{Dom}(\delta_T^{\max})$, and put $u = (\mathcal{J}_T^{\mathrm{gf}} - z)^{-1} f$. The full operator domain gives $u \in H^2$, so $\delta_T^{\max} u \in H^1$. The distributional extension of the adjoint identity gives

$$(\Delta^{(0)}_{\nabla^T} - z)\delta_T^{\max} u = \delta_T^{\max} f.$$

The right side lies in $L^2$; elliptic regularity upgrades $\delta_T^{\max} u$ to $H^2$. Applying the scalar resolvent proves (2.64). It follows that the resolvents at $z$ and $\overline{z}$ preserve $\mathscr{S}_T$. By self-adjointness they also preserve its orthogonal complement, which is the reducing property. The part of a self-adjoint operator on a reducing subspace is self-adjoint, and intersection with the full operator domain gives the second line of (2.65). The reducing projection commutes with the form functional calculus, giving the stated form domain. □

**Theorem 2.9** (Exact charged Landau negative mode)**.** *There is a smooth real $a \in \mathscr{S}_T$ satisfying*

$$\|a\|_{L^2} = 1, \qquad \mathcal{J}_T a = -\frac{4\pi}{D_T} a, \qquad \mathrm{YM}''_T\,|_{\nabla^T}(a, a) = -\frac{4\pi}{g^2_{\mathrm{YM}} D_T}. \tag{2.66}$$

*In particular, the non-flat smooth critical connection is a classical saddle for every finite $T$, not a gapped local vacuum.*

*Proof.* On the charge-two root line $\mathscr{L}_T^2$, put

$$\mathcal{D}_1 = \partial_{u_1}, \qquad \mathcal{D}_2 = \partial_{u_2} + 4\pi i u_1,$$

and define

$$\zeta_T = \frac{q_T - 6i m_T}{m_T + i L_T}, \qquad v_T = -\operatorname{Im}\zeta_T = \frac{D_T}{m_T^2 + L_T^2} > 0. \tag{2.67}$$

The sign is exact because

$$\operatorname{Im}\big((q_T - 6i m_T)(m_T - i L_T)\big) = -D_T.$$

The theta series

$$\Theta_T(u_1, u_2) = \sum_{\ell \in \mathbb{Z}} \exp\Big(-2\pi i \zeta_T (\ell + u_1)^2 + 4\pi i \ell u_2\Big) \tag{2.68}$$

converges normally with all derivatives: the absolute value of its $\ell$-th term is $e^{-2\pi v_T(\ell+u_1)^2}$. Reindexing gives

$$\Theta_T(u_1 + 1, u_2) = e^{-4\pi i u_2}\Theta_T(u_1, u_2), \qquad \Theta_T(u_1, u_2 + 1) = \Theta_T(u_1, u_2). \tag{2.69}$$

These are precisely the factors $j_n^2$, so $\Theta_T$ is a smooth section of $\mathscr{L}_T^2$, independent of $u_3, u_4$. Termwise differentiation gives

$$(\mathcal{D}_1 + \zeta_T \mathcal{D}_2)\Theta_T = 0. \tag{2.70}$$

The inverse transpose of the exact $(y_1, y_2)$-period block in (2.4) gives the physical covariant derivatives

$$\nabla^+_{y_1} = \frac{m_T \mathcal{D}_1 + q_T \mathcal{D}_2}{D_T}, \qquad \nabla^+_{y_2} = \frac{-L_T \mathcal{D}_1 + 6 m_T \mathcal{D}_2}{D_T}. \tag{2.71}$$

Therefore

$$\nabla^+_{y_1} - i\nabla^+_{y_2} = \frac{m_T + iL_T}{D_T}(\mathcal{D}_1 + \zeta_T \mathcal{D}_2), \tag{2.72}$$

and it kills $\Theta_T$. Orthogonality of the $u_2$-Fourier modes gives

$$\int_0^1 \int_0^1 |\Theta_T|^2 \, du_1 du_2 = \sum_{\ell \in \mathbb{Z}} \int_0^1 e^{-4\pi v_T (\ell + u_1)^2} \, du_1 = \frac{1}{2\sqrt{v_T}}.$$

Set

$$\phi_T = \left( \frac{2\sqrt{v_T}}{D_T} \right)^{1/2} \Theta_T. \tag{2.73}$$

Since the marked-coordinate volume element is $D_T d^4 u$, this section has unit $L^2(F_T)$-norm. Its covariant derivatives in the $x_1, x_2$ directions vanish. Furthermore, $[\mathcal{D}_1, \mathcal{D}_2] = 4\pi i$ and (2.71) imply

$$[\nabla^+_{y_1}, \nabla^+_{y_2}] = \frac{4\pi i}{D_T}.$$

Writing $\kappa_T = 4\pi / D_T$, direct multiplication yields

$$(\nabla^+_{y_1} - i\nabla^+_{y_2})^* (\nabla^+_{y_1} - i\nabla^+_{y_2}) = -\big((\nabla^+_{y_1})^2 + (\nabla^+_{y_2})^2\big) - \kappa_T.$$

Hence

$$(\nabla^+_{y_1} - i\nabla^+_{y_2})\phi_T = 0, \qquad -\big((\nabla^+_{y_1})^2 + (\nabla^+_{y_2})^2\big)\phi_T = \kappa_T \phi_T. \tag{2.74}$$

In $\mathfrak{sl}_2(\mathbb{C})$, set

$$X_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad X_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then $[H_{\mathrm{mag}}, X_+] = 2iX_+$, and the $X_+$-root line of $\operatorname{ad} \mathscr{E}_T \otimes \mathbb{C}$ is $\mathscr{L}_T^2$. Thus

$$a^+ = \phi_T X_+ (dy_1 - i\, dy_2)$$

is a global complexified adjoint-valued one-form. The first equation in (2.74) gives $d^*_{\nabla^T} a^+ = 0$.

Put $\mathcal{B}_T = 2\pi / D_T$, so that $\kappa_T = 2\mathcal{B}_T$ and $F_{y_1 y_2} = \mathcal{B}_T H_{\mathrm{mag}}$. The two nonzero components are $a^+_{y_1} = \phi_T X_+$ and $a^+_{y_2} = -i\phi_T X_+$. Substitution in (2.63), retaining the root charge two, gives

$$\begin{aligned} (\mathcal{J}_T a^+)_{y_1} &= \kappa_T \phi_T X_+ - 2[\mathcal{B}_T H_{\mathrm{mag}}, -i\phi_T X_+] = -\kappa_T \phi_T X_+, \\ (\mathcal{J}_T a^+)_{y_2} &= -i\kappa_T \phi_T X_+ - 2[-\mathcal{B}_T H_{\mathrm{mag}}, \phi_T X_+] = i\kappa_T \phi_T X_+. \end{aligned}$$

Thus $\mathcal{J}_T a^+ = -\kappa_T a^+$. For completeness, the coefficient $-2$ in (2.63) follows by expanding $F_{\nabla^T + ta} = F_{\nabla^T} + t\, d_{\nabla^T} a + \frac{1}{2} t^2 [a \wedge a]$, integrating the quadratic term by parts on the

background slice, and using the flat-base Weitzenbock identity: the Hodge Laplacian and the curvature part of the second variation each contribute $-[F_{\nu\mu}, a_\mu]$.

Finally take the real anti-Hermitian form

$$a = \frac{1}{\sqrt{2}} \left( \phi_T X_+ (dy_1 - i\, dy_2) - \overline{\phi_T} X_- (dy_1 + i\, dy_2) \right). \tag{2.75}$$

For $X(z) = zX_+ - \bar{z}X_-$, the fixed inner product gives $-\frac{1}{2}\operatorname{Tr}(X(z)^2) = |z|^2$. Therefore $|a|^2 = |\phi_T|^2$, so $\|a\|_{L^2} = 1$. Hermitian adjunction preserves the Coulomb equation and the real Jacobi operator; hence $\mathcal{J}_T a = -\kappa_T a$. On the background slice,

$$\mathrm{YM}''_T\,|_{\nabla^T}(a,a) = g_{\mathrm{YM}}^{-2} \langle a, \mathcal{J}_T a \rangle_{L^2},$$

which proves every identity in (2.66). □

## 2.5 Finite Gaussian quantization with independent cover and mesh

Let

$$\vartheta^j_{T,M_{\mathrm{cov}}} = P^{-\mathsf{T}}_{T,M_{\mathrm{cov}}} \mathbf{e}_j \in (\mathbb{R}^4)^*, \qquad 1 \le j \le 4. \tag{2.76}$$

For $0 \le p \le 4$, define the real finite cochain spaces

$$C^p_{N_{\mathrm{mesh}}}(T, M_{\mathrm{cov}}) = \mathrm{Map}\left( V_{N_{\mathrm{mesh}}}, \bigwedge^p (\mathbb{R}^4)^* \otimes \mathfrak{su}(2) \right). \tag{2.77}$$

The exterior powers carry the metric induced by the physical coframe (2.76); explicitly, for increasing multi-indices $I = (i_1, \dots, i_p)$ and $J = (j_1, \dots, j_p)$,

$$\left\langle \vartheta^{i_1} \wedge \cdots \wedge \vartheta^{i_p}, \vartheta^{j_1} \wedge \cdots \wedge \vartheta^{j_p} \right\rangle = \det\left( (G^{-1}_{T,M_{\mathrm{cov}}})_{i_a j_b} \right)_{a,b=1}^{p}. \tag{2.78}$$

Together with (2.31), this defines the pointwise tensor-product metric. The cochain inner product is

$$\langle a, b \rangle_p = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}}{N^4_{\mathrm{mesh}}} \sum_{n \in V_{N_{\mathrm{mesh}}}} \langle a(n), b(n) \rangle. \tag{2.79}$$

The symbol $(\vartheta^j)^\sharp$ denotes metric duality for this physical cotangent metric, and $\varepsilon(\vartheta^j)$ denotes left exterior multiplication by $\vartheta^j$. Let $\mathsf{T}_j a(n) = a(n + \mathbf{e}_j)$, set

$$D^+_{N_{\mathrm{mesh}},j} = N_{\mathrm{mesh}}(\mathsf{T}_j - 1), \qquad D^-_{N_{\mathrm{mesh}},j} a(n) = N_{\mathrm{mesh}}(a(n) - a(n - \mathbf{e}_j)), \tag{2.80}$$

and define

$$d_{N_{\mathrm{mesh}}} a = \sum_{j=1}^{4} \varepsilon(\vartheta^j_{T,M_{\mathrm{cov}}}) D^+_{N_{\mathrm{mesh}},j} a. \tag{2.81}$$

The commuting differences and anticommuting exterior multiplications give $d^2_{N_{\mathrm{mesh}}} = 0$. Its adjoint is

$$d^*_{N_{\mathrm{mesh}}} = -\sum_{j=1}^{4} \iota_{(\vartheta^j_{T,M_{\mathrm{cov}}})^\sharp} D^-_{N_{\mathrm{mesh}},j}. \tag{2.82}$$

Define the real reduced Coulomb space

$$\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = \ker(d^*_{N_{\mathrm{mesh}}} : C^1 \to C^0) \cap \left( \ker \Delta^{(1)}_{N_{\mathrm{mesh}}} \right)^\perp, \tag{2.83}$$

$$\Delta^{(1)}_{N_{\mathrm{mesh}}} = d_{N_{\mathrm{mesh}}} d^*_{N_{\mathrm{mesh}}} + d^*_{N_{\mathrm{mesh}}} d_{N_{\mathrm{mesh}}}. \tag{2.84}$$

On this finite-dimensional space put

$$\mathcal{L}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = g_{\mathrm{YM}}^{-2} d^*_{N_{\mathrm{mesh}}} d_{N_{\mathrm{mesh}}}. \tag{2.85}$$

**Proposition 2.10** (Exact finite Hessian spectrum)**.** *For*

$$\mathfrak{d}_{N_{\mathrm{mesh}}}(k)_j = N_{\mathrm{mesh}}\left(e^{2\pi i k_j/N_{\mathrm{mesh}}}-1\right), \qquad k\in(\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \tag{2.86}$$

*In the Fourier formulas, $\|\cdot\|$ on the complexified physical cotangent space denotes the Hermitian extension of the Euclidean norm. The positive spectrum of $\mathcal{L}$ is obtained from*

$$\lambda_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(k) = \frac{1}{g_{\mathrm{YM}}^2}\left\|P_T^{-\mathsf{T}}S_{M_{\mathrm{cov}}}^{-1}\mathfrak{d}_{N_{\mathrm{mesh}}}(k)\right\|^2, \qquad k\neq 0. \tag{2.87}$$

*Each complex Fourier fibre has transverse multiplicity $3\dim\mathfrak{su}(2)=9$. In particular,*

$$\lambda_{\min}(\mathcal{L}) = \min_{k\neq 0}\lambda_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(k) > 0. \tag{2.88}$$

*For $\mathbf{e}_1=(1,0,0,0)^{\mathsf{T}}$, the exact mode value is*

$$\lambda_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(\mathbf{e}_1) = \frac{4N_{\mathrm{mesh}}^2\sin^2(\pi/N_{\mathrm{mesh}})}{g_{\mathrm{YM}}^2}\frac{L_T^2+m_T^2}{D_T^2}; \tag{2.89}$$

*it is independent of $M_{\mathrm{cov}}$.*

*Proof.* Translation is unitary for (2.79), so $(D_j^+)^* = -D_j^-$. Exterior multiplication by $\vartheta^j$ has adjoint contraction by $(\vartheta^j)^\sharp$, proving (2.82). On the Fourier character $\chi_k(n)=e^{2\pi i k\cdot n/N_{\mathrm{mesh}}}$, the derivative symbol is exterior multiplication by

$$\zeta(k) = P_{T,M_{\mathrm{cov}}}^{-\mathsf{T}}\mathfrak{d}_{N_{\mathrm{mesh}}}(k) = P_T^{-\mathsf{T}}S_{M_{\mathrm{cov}}}^{-1}\mathfrak{d}_{N_{\mathrm{mesh}}}(k).$$

The Hermitian exterior-algebra identity

$$\iota_{\overline{\zeta}^\sharp}\varepsilon(\zeta)+\varepsilon(\zeta)\iota_{\overline{\zeta}^\sharp} = \|\zeta\|^2\,\mathrm{id}$$

shows that on the Coulomb polarization $\ker\iota_{\overline{\zeta}^\sharp}\subset(\mathbb{C}^4)^*$, the operator $d^*d$ is multiplication by $\|\zeta\|^2$. For $k\neq 0$, $\mathfrak{d}_{N_{\mathrm{mesh}}}(k)\neq 0$, so invertibility of $P_{T,M_{\mathrm{cov}}}$ gives $\zeta(k)\neq 0$. The contraction has rank one, and its kernel has dimension three. The zero Fourier fibre consists exactly of constant harmonic cochains and was removed in (2.83). Fourier completeness now proves (2.87)–(2.88).

For $k=\mathbf{e}_1$, the only nonzero component of $\mathfrak{d}_{N_{\mathrm{mesh}}}(k)$ has modulus

$$2N_{\mathrm{mesh}}\sin(\pi/N_{\mathrm{mesh}}),$$

and $S_{M_{\mathrm{cov}}}^{-1}\mathbf{e}_1=\mathbf{e}_1$. Solving $P_T^{\mathsf{T}}\xi=\mathbf{e}_1$, or directly inverting the lower $2\times 2$ block in (2.1), gives

$$\|P_T^{-\mathsf{T}}\mathbf{e}_1\|^2 = \frac{L_T^2+m_T^2}{D_T^2}.$$

Substitution proves (2.89). □

## 2.6 Native continuum comparison and the exact all-mesh gap

The cover formula (2.88) is already exact for every $M_{\mathrm{cov}}$. On the native torus, meaning $M_{\mathrm{cov}}=1$, its minimum can be evaluated completely and compared with the continuum Yang–Mills Hessian. Let $A_0$ be the product connection on $F_T\times\mathrm{SU}(2)$, and normalize

$$\mathrm{YM}_T(A) = \frac{1}{2g_{\mathrm{YM}}^2}\int_{F_T}|F_A|^2\,\mathrm{d}\,\mathrm{vol}_{F_T}\,. \tag{2.90}$$

On one-forms let $\mathrm{d}_1$ be the maximal closed exterior derivative

$$\mathrm{d}_1 : L^2\Omega^1(F_T;\mathfrak{su}(2)) \longrightarrow L^2\Omega^2(F_T;\mathfrak{su}(2)),$$

with domain

$$\mathrm{Dom}(\mathrm{d}_1) = \{a \in L^2\Omega^1 : \mathrm{d}a \in L^2\Omega^2 \text{ distributionally}\}. \tag{2.91}$$

**Proposition 2.11** (Exact native continuum Hessian and Coulomb spectrum)**.** *For smooth one-forms $a, b$,*

$$\mathrm{Hess}_{A_0}\,\mathrm{YM}_T(a,b) = g_{\mathrm{YM}}^{-2}\langle \mathrm{d}a, \mathrm{d}b\rangle_{L^2}. \tag{2.92}$$

*The associated closed form and nonnegative self-adjoint operator are*

$$\mathfrak{l}_T[a] = g_{\mathrm{YM}}^{-2}\|\mathrm{d}_1 a\|_{L^2}^2, \qquad \mathrm{Dom}(\mathfrak{l}_T) = \mathrm{Dom}(\mathrm{d}_1), \tag{2.93}$$

$$\mathcal{L}_T^{\mathrm{cont}} = g_{\mathrm{YM}}^{-2}\mathrm{d}_1^*\mathrm{d}_1, \qquad \mathrm{Dom}(\mathcal{L}_T^{\mathrm{cont}}) = \{a \in \mathrm{Dom}(\mathrm{d}_1) : \mathrm{d}_1 a \in \mathrm{Dom}(\mathrm{d}_1^*)\}. \tag{2.94}$$

*The reduced Coulomb space is the closed subspace*

$$\mathscr{C}_T^{\mathrm{cont}} = \{a \in L^2\Omega^1(F_T;\mathfrak{su}(2)) : \mathrm{d}^* a = 0 \text{ distributionally}, \quad a \perp \mathcal{H}^1(F_T;\mathfrak{su}(2))\}. \tag{2.95}$$

*It reduces $\mathcal{L}_T^{\mathrm{cont}}$. The restricted form domain is $H^1\Omega^1 \cap \mathscr{C}_T^{\mathrm{cont}}$, and the restricted operator domain is $H^2\Omega^1 \cap \mathscr{C}_T^{\mathrm{cont}}$.*

*For $k \in \mathbb{Z}^4 \setminus \{0\}$, the character $\chi_{T,k}([x]) = \exp(2\pi i\langle P_T^{-\mathsf{T}}k, x\rangle)$ has Coulomb polarization*

$$\ker \iota_{(P_T^{-\mathsf{T}}k)^\sharp} \otimes \mathfrak{su}(2)_{\mathbb{C}} \tag{2.96}$$

*of complex dimension nine, and the operator eigenvalue on that fibre is*

$$\lambda_T^{\mathrm{cont}}(k) = \frac{4\pi^2}{g_{\mathrm{YM}}^2}|P_T^{-\mathsf{T}}k|^2. \tag{2.97}$$

*For all sufficiently large $T$, put*

$$a_{\gamma,T} = \frac{L_T^2 + m_T^2}{D_T^2}, \qquad a_{u,T} = \frac{q_T^2 + 36m_T^2}{D_T^2}, \qquad I_T = \{j \in \{\gamma, u\} : a_{j,T} = \min(a_{\gamma,T}, a_{u,T})\}. \tag{2.98}$$

*Then the first reduced continuum eigenvalue and its real multiplicity are*

$$\gamma_T^{\mathrm{cont}} = \frac{4\pi^2}{g_{\mathrm{YM}}^2}\min\{a_{\gamma,T}, a_{u,T}\}, \qquad \mathrm{mult}_{\mathbb{R}}(\gamma_T^{\mathrm{cont}}) = 18\,\#I_T. \tag{2.99}$$

*Proof.* With $F_{A_0+a} = \mathrm{d}a + \frac{1}{2}[a \wedge a]$, the coefficient of $st$ in $|F_{A_0+sa+tb}|^2$ is $2\langle \mathrm{d}a, \mathrm{d}b\rangle$, which proves (2.92). The maximal distributional derivative is closed and densely defined. The first representation theorem for its graph-norm square therefore gives (2.93)–(2.94) with exactly the displayed domains.

On the flat torus the Fourier identities are

$$\begin{aligned}
\mathrm{d}(\chi_{T,k}\alpha) &= 2\pi i\chi_{T,k}\varepsilon(\xi_{T,k})\alpha,\\
\mathrm{d}^*(\chi_{T,k}\alpha) &= -2\pi i\chi_{T,k}\iota_{\xi_{T,k}^\sharp}\alpha,\\
\mathrm{d}^*\mathrm{d}(\chi_{T,k}\alpha) &= 4\pi^2\chi_{T,k}\big(|\xi_{T,k}|^2\alpha - \varepsilon(\xi_{T,k})\iota_{\xi_{T,k}^\sharp}\alpha\big), \qquad \xi_{T,k} = P_T^{-\mathsf{T}}k.
\end{aligned}$$

They show directly that $\mathscr{C}_T^{\mathrm{cont}}$ reduces the operator. They also identify its form and operator domains respectively with $H^1 \cap \mathscr{C}_T^{\mathrm{cont}}$ and $H^2 \cap \mathscr{C}_T^{\mathrm{cont}}$. For $k \neq 0$, contraction

by $\xi^{\sharp}_{T,k}$ has rank one on the four-dimensional complexified cotangent space. Its kernel therefore has dimension three; tensoring with $\mathfrak{su}(2)_{\mathbb{C}}$ gives dimension nine, and the last displayed identity proves (2.97). The zero mode is precisely the constant harmonic sector removed in (2.95).

It remains to identify the shortest nonzero dual vectors. Write $k = (r, s, n_1, n_2)^{\mathsf{T}}$. Solving $P_T^{\mathsf{T}} x = k$, the third and fourth equations give $x_1 = n_1$, $x_3 = n_2$. Thus $|P_T^{-\mathsf{T}} k| \geq 1$ when $(n_1, n_2) \neq (0, 0)$. When those two indices vanish,

$$P_T^{-\mathsf{T}}(r, s, 0, 0)^{\mathsf{T}} = \left(0, \frac{m_T r + q_T s}{D_T}, 0, \frac{-L_T r + 6 m_T s}{D_T}\right). \tag{2.100}$$

The scaled two-dimensional matrix in this expression satisfies

$$\frac{T}{D_T}\begin{pmatrix} m_T & q_T \\ -L_T & 6m_T \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + O_{\mathrm{op}}(T^{-1}).$$

Hence, uniformly in real $(r, s)$,

$$(1 - C/T)\sqrt{r^2 + s^2} \leq T|P_T^{-\mathsf{T}}(r, s, 0, 0)^{\mathsf{T}}| \leq (1 + C/T)\sqrt{r^2 + s^2}.$$

For large $T$, every integer pair of squared norm at least two is longer than each unit pair, while every mode with $(n_1, n_2) \neq 0$ has length at least one. The only possible shortest vectors are therefore the two signed pairs with indices $\pm(1, 0, 0, 0)$ and $\pm(0, 1, 0, 0)$. Their squared lengths, read from (2.100), are precisely $a_{\gamma,T}$ and $a_{u,T}$. Each signed pair contributes eighteen real dimensions, proving (2.99). □

Set

$$\rho_N = 4N^2 \sin^2(\pi/N), \qquad \eta_N = \begin{cases} 1, & N = 2, \\ 2, & N > 2. \end{cases} \tag{2.101}$$

**Theorem 2.12** (Exact native finite gap in every mesh regime)**.** *There is $T_1$, independent of $N \geq 2$, such that for $T \geq T_1$, the first eigenvalue of $\mathcal{L}_{T,1,N}$ on $\mathscr{C}_{T,1,N}$ is*

$$\gamma^{\mathrm{fin}}_{N,T} = \frac{\rho_N}{g^2_{\mathrm{YM}}} \min\{a_{\gamma,T}, a_{u,T}\} = \frac{\rho_N}{4\pi^2}\gamma^{\mathrm{cont}}_T. \tag{2.102}$$

*Its real multiplicity is exactly*

$$\mathrm{mult}_{\mathbb{R}}(\gamma^{\mathrm{fin}}_{N,T}) = 9\eta_N \, \#I_T. \tag{2.103}$$

*Uniformly for every integer-valued choice $N(T) \geq 2$, there are constants $0 < c < C < \infty$ such that*

$$\frac{c}{T^2} \leq \gamma^{\mathrm{fin}}_{N(T),T} \leq \frac{C}{T^2} \tag{2.104}$$

*for all large $T$. If $N(T) \to \infty$, then*

$$\gamma^{\mathrm{fin}}_{N(T),T} = \frac{4\pi^2}{g^2_{\mathrm{YM}} T^2}\left(1 + O(T^{-1}) + O(N(T)^{-2})\right). \tag{2.105}$$

*The largest physical edge in the native cubulation is*

$$\max_{1 \leq j \leq 4} \frac{|P_T \mathbf{e}_j|}{N(T)} = \frac{T}{N(T)}(1 + O(T^{-1})), \tag{2.106}$$

*so this varying continuum torus is resolved exactly when $N(T)/T \to \infty$. This geometric condition is not needed for the algebraic gap bounds above.*

*Proof.* Put $\sigma_N = 2N\sin(\pi/N)$, so $\rho_N = \sigma_N^2$. The least nonzero modulus of a component of $\mathfrak{d}_N(k)$ is $\sigma_N$, attained exactly at frequency $\pm 1$, with the two signs equal when $N = 2$. If $\mathfrak{d}_N(k)$ is supported in its first two components but $k \notin \{\pm\mathbf{e}_1, \pm\mathbf{e}_2\}$, then

$$|\mathfrak{d}_N(k)_1|^2 + |\mathfrak{d}_N(k)_2|^2 \geq 2\sigma_N^2. \tag{2.107}$$

Indeed, either both components are nonzero, or the single nonzero frequency is not $\pm 1$; in the latter case $\sin(2\pi/N)/\sin(\pi/N) = 2\cos(\pi/N) \geq \sqrt{2}$ for $N \geq 4$, and $N = 2, 3$ are immediate.

On the first two components, (2.100) and the uniform singular-value estimate in its proof show that the factor $\sqrt{2}$ in (2.107) separates every other frequency from the two coordinate frequencies, uniformly in $N$, once $T$ is large. For a general complex vector $z \in \mathbb{C}^4$, the third and fourth equations in $P_T^{\mathsf{T}} x = z$ give $x_1 = z_3$, $x_3 = z_4$. Thus any finite frequency with nonzero third or fourth component satisfies

$$|P_T^{-\mathsf{T}}\mathfrak{d}_N(k)|^2 \geq |\mathfrak{d}_N(k)_3|^2 + |\mathfrak{d}_N(k)_4|^2 \geq \sigma_N^2,$$

whereas either coordinate candidate is $O(\sigma_N^2/T^2)$. Therefore the only minimizers are the signed coordinate pairs selected by $I_T$. The exact mode formula (2.87) gives (2.102). A signed complex Fourier fibre has dimension nine; reality pairs the two signs when $N > 2$, while they coincide when $N = 2$. This proves (2.103).

Finally, the standing assumptions give $a_{\gamma,T} = T^{-2}(1 + O(T^{-1}))$ and $a_{u,T} = T^{-2}(1 + O(T^{-1}))$. The elementary sine bounds yield $16 \leq \rho_N \leq 4\pi^2$, proving (2.104). When $N(T) \to \infty$, $\rho_{N(T)} = 4\pi^2(1 + O(N(T)^{-2}))$, which proves (2.105). The first two columns of $P_T$ have norm $T(1 + O(T^{-1}))$, while its last two columns have norm one. Division by $N(T)$ proves (2.106) and the final assertion. □

**Remark 2.13** (Limits of the comparison)**.** The exact minimum in Theorem 2.12 is a statement only for $M_{\mathrm{cov}} = 1$. For a nontrivial cover, $S_{M_{\mathrm{cov}}}^{-1}$ creates additional frequency scales, so the general statement remains the exact minimization (2.88) together with proved mode bounds such as (2.89). Moreover, $\mathcal{L}_T^{\mathrm{cont}}$ is a classical Hessian operator; no quantization or unitary identification with the finite Gaussian Hilbert space is contained in Proposition 2.11.

Let $da$ denote Lebesgue measure on $\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}$ induced by (2.79) and define

$$\mathcal{H}^{\mathrm{Gauss}}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = L^2(\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}, da). \tag{2.108}$$

Put $r = \dim_{\mathbb{R}} \mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}$. In any orthonormal linear coordinates $(x_1, \dots, x_r)$ for the inner product induced by (2.79), define

$$\Delta_{\mathscr{C}} = \sum_{s=1}^{r} \frac{\partial^2}{\partial x_s^2}. \tag{2.109}$$

This definition is independent of the chosen orthonormal coordinates.

**Theorem 2.14** (Finite Gaussian oscillator and a resolved cusp sequence)**.** *On the initial domain $C_c^\infty(\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}; \mathbb{C})$, consider the symmetric quadratic differential operator*

$$\widehat{H}^{\mathrm{Gauss}}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = -\frac{1}{2}\Delta_{\mathscr{C}} + \frac{1}{2}\langle a, \mathcal{L}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} a\rangle \tag{2.110}$$

*where the second summand denotes multiplication by the displayed quadratic function of $a$. Its Friedrichs realization on (2.108) has a unique Gaussian ground state and spectral gap*

$$\operatorname{gap}(\widehat{H}^{\mathrm{Gauss}}) = \sqrt{\lambda_{\min}(\mathcal{L}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}})}. \tag{2.111}$$

*Consequently,*

$$\operatorname{gap}(\widehat{H}^{\mathrm{Gauss}}) \le \frac{2N_{\mathrm{mesh}}\sin(\pi/N_{\mathrm{mesh}})}{g_{\mathrm{YM}}}\frac{\sqrt{L_T^2+m_T^2}}{D_T}. \tag{2.112}$$

*For $M_{\mathrm{cov}} = 1$ and all sufficiently large $T$, the exact native formula is*

$$\operatorname{gap}(\widehat{H}^{\mathrm{Gauss}}_{T,1,N_{\mathrm{mesh}}}) = \frac{2N_{\mathrm{mesh}}\sin(\pi/N_{\mathrm{mesh}})}{g_{\mathrm{YM}}}\sqrt{\min\{a_{\gamma,T}, a_{u,T}\}}. \tag{2.113}$$

*The complex dimension of the first-excited eigenspace is $9\eta_{N_{\mathrm{mesh}}}\#I_T$. Uniformly for every $N_{\mathrm{mesh}}(T) \ge 2$, this gap is comparable to $T^{-1}$; if $N_{\mathrm{mesh}}(T) \to \infty$, it equals*

$$\frac{2\pi}{g_{\mathrm{YM}}T}\left(1 + O(T^{-1}) + O(N_{\mathrm{mesh}}(T)^{-2})\right). \tag{2.114}$$

*There is an explicit diagonal sequence for which the cover geometry tends locally to $\mathbb{R}^4$, the cubical mesh resolves that geometry, and the Gaussian gap tends to zero. Explicitly, for integers $j \to \infty$, take*

$$T_j = j^2, \qquad M_{\mathrm{cov},j} = j, \qquad N_{\mathrm{mesh},j} = j^3. \tag{2.115}$$

*Then*

$$\begin{aligned} \operatorname{sys}(\widetilde{F}_{T_j,M_{\mathrm{cov},j}}) &\longrightarrow \infty, \\ \max_r \frac{\|P_{T_j,M_{\mathrm{cov},j}}\mathbf{e}_r\|}{N_{\mathrm{mesh},j}} &\longrightarrow 0, \\ \operatorname{gap}(\widehat{H}^{\mathrm{Gauss}}) &\longrightarrow 0, \\ S^{\log}_{T_j,M_{\mathrm{cov},j},N_{\mathrm{mesh},j}}(U) &\longrightarrow 0. \end{aligned} \tag{2.116}$$

*Thus the pointed cover tori converge locally isometrically to $\mathbb{R}^4$ while their finite linearized Gaussian systems are simultaneously refined.*

*Proof.* The real finite-dimensional operator $\mathcal{L}$ is positive definite on the reduced Coulomb space. Choose an orthonormal eigenbasis with eigenvalues $0 < \lambda_1 \le \cdots \le \lambda_r$. In the resulting coordinates, (2.110) is the tensor sum

$$\sum_{s=1}^{r}\left(-\frac{1}{2}\frac{\partial^2}{\partial x_s^2} + \frac{1}{2}\lambda_s x_s^2\right).$$

The normalized Hermite basis has eigenvalues $\sum_s (n_s + \frac{1}{2})\sqrt{\lambda_s}$, with $n_s \in \mathbb{Z}_{\ge 0}$. The all-zero multi-index is the unique ground state, and the smallest possible positive excitation is $\sqrt{\lambda_1}$. This proves (2.111); applying the minimum to the explicit $\mathbf{e}_1$-mode in (2.89) proves (2.112). When $M_{\mathrm{cov}} = 1$, Theorem 2.12 evaluates $\lambda_1$ and its real eigenspace exactly. Taking its positive square root proves (2.113); the same operation on (2.104) and (2.105) gives the stated comparison and (2.114).

Since $T_j, M_{\mathrm{cov},j} \to \infty$, Proposition 2.3 proves both the systole conclusion and the pointed local-isometric $\mathbb{R}^4$ limit.

The first two columns of $P_T$ have norm $O(T)$, while the last two columns of $P_{T,M_{\mathrm{cov}}}$ have norm exactly $M_{\mathrm{cov}}$. Therefore the largest physical edge in the $N_{\mathrm{mesh}}$-subdivision has length $O(\max\{T, M_{\mathrm{cov}}\}/N_{\mathrm{mesh}})$, which is $O(j^{-1})$ for (2.115). Finally, (2.3) gives $\sqrt{L_T^2 + m_T^2}/D_T = T^{-1}(1 + O(T^{-1}))$, while $2N\sin(\pi/N) \le 2\pi$. The right-hand side of (2.112) is therefore $O(j^{-2})$. Moreover, (2.40) is $O(j^{-2})$, since $M_{\mathrm{cov},j}^2/D_{T_j} = j^2/(j^4+O(j^2))$. This proves every assertion in (2.116). □

## 2.7 A gapless auxiliary direct-integral operator

The closing of the native scalar eigenvalues can be attached to one operator without identifying that operator with a physical Hamiltonian. Use the coordinate diffeomorphism $[y] \mapsto [P_T y]$ from (2.20); normalized Haar measure on $F_T$ pulls back to normalized Lebesgue measure $dy$ on $\mathbb{T}^4 = \mathbb{R}^4/\mathbb{Z}^4$. For $k \in \mathbb{Z}^4 \setminus \{0\}$, put

$$e_k(y) = e^{2\pi i k\cdot y}, \qquad \lambda_k(T) = 4\pi^2 |P_T^{-\mathsf{T}} k|^2.$$

Fix $T_1 \ge T_0$ sufficiently large for the estimates following (2.100), and define

$$\mathscr{H}_{\mathrm{aux}} = L^2\Big([T_1,\infty), dT; L_0^2(\mathbb{T}^4, dy)\Big) \cong \bigoplus_{k\in\mathbb{Z}^4\setminus\{0\}} L^2([T_1,\infty), dT), \tag{2.117}$$

$$(\mathscr{L}_{\mathrm{aux}} u)_k(T) = \lambda_k(T) u_k(T), \tag{2.118}$$

$$\mathrm{Dom}(\mathscr{L}_{\mathrm{aux}}) = \left\{ u \in \mathscr{H}_{\mathrm{aux}} : \sum_{k\neq 0} \int_{T_1}^{\infty} \lambda_k(T)^2 |u_k(T)|^2 \, dT < \infty \right\}. \tag{2.119}$$

Here $L_0^2$ is the orthogonal complement of the constants. Thus $\mathscr{L}_{\mathrm{aux}}$ is, by definition, the direct integral of the mean-zero scalar Laplacians on the native fibres, expressed on a fixed Hilbert space.

**Theorem 2.15** (Exact zero essential threshold)**.** *The operator $\mathscr{L}_{\mathrm{aux}}$ is nonnegative and self-adjoint on the domain (2.119). Its kernel is zero, but*

$$0 \in \mathrm{Spec}_{\mathrm{ess}}(\mathscr{L}_{\mathrm{aux}}). \tag{2.120}$$

*It therefore has no positive spectral threshold above zero.*

*Proof.* For each $k \neq 0$, $\lambda_k$ is a positive continuous real function. Multiplication by $\lambda_k$ on

$$\{v \in L^2([T_1,\infty)) : \lambda_k v \in L^2([T_1,\infty))\}$$

is nonnegative and self-adjoint: testing the adjoint against bounded compactly supported functions forces $\lambda_k v \in L^2$, and the adjoint acts by the same real multiplier. The orthogonal sum of these multiplication operators has exactly the domain (2.119), proving self-adjointness. Strict positivity of every $\lambda_k(T)$ proves that the kernel is zero.

Take $k_0 = (1,0,0,0)$. Formula (2.100) and the standing cusp bounds give

$$\lambda_{k_0}(T) = 4\pi^2 \frac{L_T^2 + m_T^2}{D_T^2} = \frac{4\pi^2}{T^2}(1 + O(T^{-1})). \tag{2.121}$$

For integers $n > T_1$, let $u^{(n)}$ have only its $k_0$-component nonzero, with

$$u_{k_0}^{(n)}(T) = \mathbf{1}_{[n,n+1]}(T).$$

The $u^{(n)}$ are orthonormal members of $\mathrm{Dom}(\mathscr{L}_{\mathrm{aux}})$, converge weakly to zero, and satisfy

$$\|\mathscr{L}_{\mathrm{aux}} u^{(n)}\|^2 = \int_n^{n+1} \lambda_{k_0}(T)^2 \, dT \le \sup_{T\in[n,n+1]} \lambda_{k_0}(T)^2 \longrightarrow 0.$$

This is a Weyl singular sequence at zero. Because the kernel is zero, it places zero in the essential rather than discrete spectrum, proving (2.120). □

This theorem concerns an auxiliary scalar direct integral whose base measure is $dT$. It is neither the finite Gaussian oscillator nor a Yang–Mills Hamiltonian, and it supplies no map between their Hilbert spaces.

## 2.8 The exponential map and the exact physical boundary

Let $C^1 = C^1_{N_{\mathrm{mesh}}}(T, M_{\mathrm{cov}})$, regarded only as a finite-dimensional Lie-algebra cochain space, and define componentwise

$$\mathrm{Exp}_{N_{\mathrm{mesh}}} : C^1 \longrightarrow \mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, \qquad a \longmapsto (\exp(a_e/N_{\mathrm{mesh}}))_e \,. \tag{2.122}$$

Here, if $a(n) = \sum_{j=1}^4 \vartheta^j_{T,M_{\mathrm{cov}}} \otimes a_j(n)$, then $a_e = a_j(n)$ for the oriented edge $e = (n, j)$.

**Proposition 2.16** (Local chart, not a global embedding)**.** *The map* (2.122) *is smooth and a local diffeomorphism at zero, but it is not globally injective. Under left trivialization of every tangent space at the identity,*

$$D\,\mathrm{Exp}_{N_{\mathrm{mesh}}}(0)\, a = N_{\mathrm{mesh}}^{-1} a. \tag{2.123}$$

*If* $S_W : \mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}} \to \mathbb{R}$ *is any smooth compact-link action for which the identity configuration* $\mathbf{1} = (1)_{e\in E_{N_{\mathrm{mesh}}}}$ *is critical, then*

$$D^2(S_W \circ \mathrm{Exp}_{N_{\mathrm{mesh}}})_0[a,b] = N_{\mathrm{mesh}}^{-2} D^2(S_W)_{\mathbf{1}}[a,b] \tag{2.124}$$

*under the same tangent identification. Thus, for a prescribed cochain quadratic form* $Q$*,*

$$D^2(S_W \circ \mathrm{Exp}_{N_{\mathrm{mesh}}})_0 = Q$$

*holds exactly when*

$$D^2(S_W)_{\mathbf{1}} = N_{\mathrm{mesh}}^2 Q$$

*under the coefficient identification. Both this scaling and the tensor structure must be verified for the chosen compact-link action; neither follows from the exponential map alone.*

*Proof.* For the Lie-group exponential, $D\exp_0$ is the identity. Applying the chain rule to $a_e \mapsto a_e/N_{\mathrm{mesh}}$ gives (2.123), which is invertible, so the inverse function theorem gives a local diffeomorphism. It is not globally injective: a cochain which is zero on every edge except one, where it equals $2\pi N_{\mathrm{mesh}} H_{\mathrm{link}}$, has the same image as the zero cochain by (2.32). Finally, the second-derivative chain rule contains a term involving $DS_W(\mathbf{1})$, which vanishes by criticality; the two remaining first derivatives of Exp each supply the factor $N_{\mathrm{mesh}}^{-1}$. This proves (2.124). □

The Gaussian Hilbert space in (2.108) is not the compact-link Yang–Mills Hilbert space. At this finite lattice, the latter kinematic and gauge-invariant spaces would be

$$\mathcal{H}^{\mathrm{link}}_{N_{\mathrm{mesh}}} = L^2(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, dU), \qquad \mathcal{H}^{\mathrm{inv}}_{N_{\mathrm{mesh}}} = \left(\mathcal{H}^{\mathrm{link}}_{N_{\mathrm{mesh}}}\right)^{\mathrm{SU}(2)^{V_{N_{\mathrm{mesh}}}}}, \tag{2.125}$$

where $dU$ is product normalized Haar measure and the superscript denotes the nonlinear vertex-gauge action. Here "gauge-invariant" refers only to the link-configuration space of this four-dimensional cell complex. Without a chosen time direction or transfer matrix, it is not asserted to be a canonical Hamiltonian physical Hilbert space. Lüscher's construction of a selfadjoint, strictly positive transfer matrix [2] and Menotti–Pelissetto's reflection-positivity theorem for the Wilson action [3] concern the standard Wilson lattice setting after the requisite reflection/time structure has been fixed; they do not supply those omitted choices for the nonorthogonal sector functional (2.41). The Osterwalder–Schrader reconstruction boundary is likewise the one stated in the primary Euclidean axioms papers [4, 5], not a consequence of a finite Gaussian oscillator calculation. The differences are structural:

1. $\mathcal{H}^{\mathrm{Gauss}}$ is an $L^2$-space on a noncompact linear Coulomb slice with Lebesgue measure; $\mathcal{H}^{\mathrm{inv}}$ is a gauge-invariant subspace of an $L^2$-space on compact link variables with Haar measure.

2. The local exponential map is neither a global coordinate chart nor a measure-preserving bijection. No unitary map between the two displayed Hilbert spaces has been constructed.

3. A link configuration $U$ determines the Dirac distribution $\delta_U$, but

$$\delta_U \in \mathcal{D}'(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}), \qquad \delta_U \notin L^2(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, dU). \tag{2.126}$$

   Indeed, product Haar measure is nonatomic, so a point mass is singular with respect to it and has no square-integrable density.

4. No nonlinear compact-link Hamiltonian, Gauss-law spectral problem, time direction, reflection-positive transfer matrix, renormalized-scale limit, or Osterwalder–Schrader reconstruction is supplied here.

Accordingly, (2.111) is an exact theorem about the finite quadratic oscillator defined by the Coulomb-restricted operator $g_{\mathrm{YM}}^{-2} d^* d$. It is not an eigenvalue statement about a nonperturbative physical Yang–Mills Hamiltonian, and the small classical action in (2.40) is not by itself a physical-state Rayleigh quotient. Similarly, $\widetilde{\Delta}_{T,M_{\mathrm{cov}}}$ is a scalar geometric Laplacian and $\mathcal{L}_T^{\mathrm{cont}}$ is a classical Yang–Mills Hessian at the product connection. No common direct-integral operator, physical Hamiltonian, or map from either of those operators to the compact-link Hilbert space is asserted.

## 2.9 Wilson-loop states, entanglement, and the exact coercive bridge

The preceding distinction can be made by exact countermodels, rather than by terminology. Put

$$\mathscr{X}_N = \mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, \qquad \mathscr{G}_N = \mathrm{SU}(2)^{V_{N_{\mathrm{mesh}}}},$$

and use the vertex-gauge action compatible with (2.30):

$$(g \cdot U)_{n,j} = g(n)^{-1} U_{n,j} g(n + \mathbf{e}_j). \tag{2.127}$$

If $C = (e_1^{\varepsilon_1}, \dots, e_r^{\varepsilon_r})$ is an oriented closed edge path, with $\varepsilon_k \in \{1, -1\}$, let

$$W_C(U) = \frac{1}{2} \operatorname{Re} \operatorname{tr} \left( U_{e_1}^{\varepsilon_1} \cdots U_{e_r}^{\varepsilon_r} \right). \tag{2.128}$$

Here an edge with exponent $-1$ is traversed opposite to its chosen orientation. Under (2.127), the ordered product around $C$ is conjugated at its base vertex, so $W_C$ is gauge invariant. A *Euclidean link state* in this paragraph means a $\mathscr{G}_N$-invariant Borel probability measure on $\mathscr{X}_N$. This is deliberately not called a Hamiltonian state: no time direction or transfer matrix has been chosen.

**Proposition 2.17** (Link kinematics and one magnetic field select no area law)**.** *There are $\mathscr{G}_N$-invariant Borel probability measures $\mu_{\mathrm{pg}}$ and $\mu_{\mathrm{H}}$ on the same configuration space $\mathscr{X}_N$ such that*

$$\int_{\mathscr{X}_N} W_C \, d\mu_{\mathrm{pg}} = 1 \qquad \text{for every closed edge path } C, \tag{2.129}$$

$$\int_{\mathscr{X}_N} W_C \, d\mu_{\mathrm{H}} = 0 \qquad \text{if some unoriented edge occurs exactly once in } C. \tag{2.130}$$

*Consequently, for each $0 \le t \le 1$, the invariant state $\mu_t = t\mu_{\mathrm{pg}} + (1-t)\mu_{\mathrm{H}}$ has expectation $t$ for every path covered by the second line. Thus the configuration space, gauge action, and existence of the explicit flux in Theorem 2.6 do not determine Wilson-loop decay or a string tension.*

*Proof.* Let $dg_V$ be normalized product Haar measure on $\mathscr{G}_N$, and let $\mu_{\mathrm{pg}}$ be its push-forward under

$$\mathscr{G}_N \longrightarrow \mathscr{X}_N, \qquad g \longmapsto g \cdot \mathbf{1}.$$

Bi-invariance of $dg_V$ makes this push-forward gauge invariant: applying a fixed gauge transformation replaces $g$ by its pointwise right translate. Along a positively oriented edge $v \to w$, the corresponding link is $g(v)^{-1}g(w)$; the inverse occurs for the opposite orientation. The factors telescope around every closed path, giving the identity matrix and proving (2.129).

Let $\mu_{\mathrm{H}}$ be normalized product Haar measure on $\mathscr{X}_N$. Every fixed gauge transformation acts on each link by a left and a right translation, so bi-invariance of Haar measure proves gauge invariance. Suppose an unoriented edge $e$ occurs exactly once in $C$. After conditioning on all other links, the matrix inside the trace in (2.128) is of the form $AU_eB$ or $AU_e^{-1}B$, with $A, B$ fixed. In the defining representation,

$$\int_{\mathrm{SU}(2)} U \, dU = 0, \qquad \int_{\mathrm{SU}(2)} U^{-1} \, dU = 0 :$$

left translation by the central element $-I$ changes either integral to its negative. Fubini's theorem proves (2.130). Linearity of integration gives the assertion for $\mu_t$. All these measures exist independently of the singled-out point of $\mathscr{X}_N$ defined in (2.33)–(2.35). □

The word *entangled* also requires a specified tensor product. For complex Hilbert spaces $\mathfrak{h}_A, \mathfrak{h}_B$ of dimension at least two, a nonzero vector of $\mathfrak{h}_A \otimes \mathfrak{h}_B$ is a product vector if it equals $a \otimes b$; otherwise it is entangled.

**Proposition 2.18** (Entanglement is neither a sector nor an energy floor)**.** *The entangled vectors, together with zero, do not form a linear subspace. They are norm dense and have dense linear span, so no orthogonal projection has precisely the entangled vectors as its range. Moreover:*

1. *on $\mathbb{C}^2 \otimes \mathbb{C}^2$, the bounded nonnegative self-adjoint operator*

$$H_B = I - |\Omega_B\rangle\langle\Omega_B|, \qquad \Omega_B = 2^{-1/2}(|00\rangle + |11\rangle),$$

*with domain the whole tensor product, has the entangled vector $\Omega_B$ as its unique zero-energy ground state and has gap one;*

2. *there is a bounded nonnegative self-adjoint operator, again with full domain, which has a unique entangled zero-energy ground state and entangled positive-energy eigenvectors whose eigenvalues converge to zero.*

*Hence a statement that some degrees of freedom must be entangled supplies neither a positive energy nor a positive spectral gap.*

*Proof.* Choose orthonormal $e_0, e_1 \in \mathfrak{h}_A$ and $f_0, f_1 \in \mathfrak{h}_B$. The Bell vectors

$$\Psi_\pm = 2^{-1/2}(e_0 \otimes f_0 \pm e_1 \otimes f_1)$$

have coefficient-matrix rank two and are entangled, whereas their sum is the product vector $\sqrt{2}\, e_0 \otimes f_0$. Thus the set is not linear. Any product vector $a \otimes b$ is the norm limit, as $\varepsilon \to 0$, of $a \otimes b + \varepsilon a' \otimes b'$, where $a' \perp a$ and $b' \perp b$; for $\varepsilon \neq 0$ the latter has coefficient rank two. This proves density. Also

$$a \otimes b = \frac{1}{2}\big[(a \otimes b + a' \otimes b') + (a \otimes b - a' \otimes b')\big],$$

so product vectors, which span a dense subspace, lie in the span of entangled vectors. A projection range is a closed linear subspace.

For the first operator assertion, the rank-one term is the orthogonal projection onto the Bell line. Hence $H_B$ has spectrum $\{0, 1\}$ and kernel $\mathbb{C}\Omega_B$.

For the second, take $\mathfrak{h}_A = \mathfrak{h}_B = \ell^2(\mathbb{N}_0)$, with orthonormal bases $(e_j)$ and $(f_j)$, and set

$$\Omega = 2^{-1/2}(e_0 \otimes f_0 + e_1 \otimes f_1), \qquad \Phi_n = 2^{-1/2}(e_{2n} \otimes f_{2n+1} + e_{2n+1} \otimes f_{2n}) \quad (n \geq 1).$$

These vectors are orthonormal. Define $H$, on the whole tensor product, by

$$H\Omega = 0, \qquad H\Phi_n = n^{-1}\Phi_n, \qquad H = I \quad \text{on } \{\Omega, \Phi_1, \Phi_2, \ldots\}^\perp. \tag{2.131}$$

This real diagonal prescription defines a bounded nonnegative self-adjoint operator. Only $\Omega$ has eigenvalue zero. Each $\Phi_n$ is entangled and has eigenvalue $1/n$, which tends to zero. More strongly, every nonzero vector in $\overline{\operatorname{span}}\{\Phi_n : n \geq 1\}$ is entangled: its coefficient operator has a nonzero rank-two block $2^{-1/2}c_n \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ for some $n$. □

The same logical obstruction occurs even without a tensor factorization. On $\ell^2(\mathbb{N}_0)$, let the Gaussian number operator be $N|n\rangle = n|n\rangle$ on $\mathrm{Dom}(N) = \{(c_n) : \sum n^2|c_n|^2 < \infty\}$. It has gap one. The bounded nonnegative self-adjoint operator $F(N)$, defined on the whole space by

$$F(N)|0\rangle = 0, \qquad F(N)|1\rangle = |1\rangle, \qquad F(N)|n\rangle = n^{-1}|n\rangle \quad (n \geq 2), \tag{2.132}$$

agrees with $N$ on the vacuum and the entire one-particle sector, yet has unique vacuum and zero excitation gap. This follows directly from the displayed eigenvalues. Thus even exact vacuum and one-particle Gaussian data do not determine a nonlinear completion.

We finally record the positive statement that identifies the missing map. Let $A_{\mathrm{phys}}$ be a lower-semibounded self-adjoint operator on a Hilbert space $\mathcal{H}_{\mathrm{phys}}$, with ground energy $E_0$, nonzero ground projection

$$P_0 = \mathbf{1}_{\{E_0\}}(A_{\mathrm{phys}}), \qquad \mathcal{H}_{\mathrm{exc}} = (I - P_0)\mathcal{H}_{\mathrm{phys}},$$

closed form $\mathfrak{a}_{\mathrm{phys}}$, and form domain $\mathcal{Q}(A_{\mathrm{phys}})$. Let $B \geq 0$ be an auxiliary self-adjoint operator with closed form $\mathfrak{b}$, form domain $\mathcal{Q}(B)$, and $\mathfrak{b}[v] \geq \gamma\|v\|^2$ for some $\gamma > 0$.

**Theorem 2.19** (Full-domain coercive gap transfer)**.** *Suppose a linear map*

$$R : \mathcal{Q}(A_{\mathrm{phys}}) \cap \mathcal{H}_{\mathrm{exc}} \longrightarrow \mathcal{Q}(B)$$

*is defined on the entire physical excitation form domain and, for constants $c, C > 0$, satisfies*

$$\|Ru\| \geq c\|u\|, \qquad \mathfrak{b}[Ru] \leq C\big(\mathfrak{a}_{\mathrm{phys}}[u] - E_0\|u\|^2\big). \tag{2.133}$$

*Then*

$$\inf \mathrm{Spec}\,(A_{\mathrm{phys}}|_{\mathcal{H}_{\mathrm{exc}}}) - E_0 \geq \frac{\gamma c^2}{C} > 0. \tag{2.134}$$

*By contrast, an isometric intertwining of $B$ with only one reducing subspace of $A_{\mathrm{phys}}$ does not imply a full gap.*

*Proof.* For every $u$ in the stated physical form domain,

$$C\big(\mathfrak{a}_{\mathrm{phys}}[u] - E_0\|u\|^2\big) \geq \mathfrak{b}[Ru] \geq \gamma\|Ru\|^2 \geq \gamma c^2\|u\|^2.$$

The ground spectral projection reduces $A_{\mathrm{phys}}$. The spectral theorem, applied to its excitation part, identifies the infimum of the Rayleigh quotient of the shifted closed form with the left-hand side of (2.134), proving the bound.

For the final assertion, take $B = I$ on $\mathbb{C}$, and on

$$\mathcal{H} = \mathbb{C}\Omega \oplus \mathbb{C} \oplus L^2((0,1), dx)$$

take the bounded nonnegative self-adjoint operator

$$A = 0 \oplus I \oplus M_x,$$

where $(M_x f)(x) = x f(x)$. The isometry $Jz = 0 \oplus z \oplus 0$ obeys $AJ = JB$, and its range is reducing and gapped. Nevertheless $A$ has unique vacuum $\Omega$, while normalized functions supported in $(0, 1/n)$ have Rayleigh quotient at most $1/n$. Thus the full excitation gap is zero. □

In the present paper, the auxiliary Gaussian oscillator is explicit, but no map $R$ satisfying (2.133) from a physical compact-link excitation form domain has been constructed. Moreover, its own gap tends to zero along (2.115); hence it cannot provide a uniform positive $\gamma$ on that sequence. The Hilbert space (2.125) also contains link variables only: no fundamental-matter Hilbert factor, pair-creation operator, hopping term, or matter mass term has been defined. String breaking by dynamical matter is therefore not an observable or spectral process typed by this model.

# 3 Conclusion

The constructions above give an exact compatibility diagram at the level that has actually been proved:

$$\{\text{smooth connections on } \widetilde{F}_{T,M_{\mathrm{cov}}}\} \xrightarrow{\mathrm{Hol}} \mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}},$$
$$\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} \xrightarrow[\text{locally}]{\mathrm{Exp}_{N_{\mathrm{mesh}}}} \mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}.$$

The first arrow is globally defined but is not claimed to be surjective. The second is a local coordinate map at zero, not a global embedding and not a measure-preserving identification. On the compact-link side, the periodic flux and both displayed action values are exact. On the linear side, the scalar cover spectrum, deck-invariant sector, continuum native Hessian spectrum, finite Fourier eigenvalues, and oscillator gap are exact. On the distinct smooth native side, the unit-flux line bundle, the reducible SU(2) Yang–Mills critical connection, and its action $2\pi^2/(g_{\mathrm{YM}}^2 D_T)$ are exact; the explicit charge-two theta section produces the reduced Jacobi eigenvalue $-4\pi/D_T$. Thus that smooth magnetic representative is a saddle at every finite $T$, and its numerical action agreement with the native link field does not identify the two configurations. The native scalar direct integral is also an exact self-adjoint operator with trivial kernel and zero in its essential spectrum. The native torus tends pointedly to $\mathbb{T}^2 \times \mathbb{R}^2$, whereas every sequence with $T_j \to \infty$ and $M_{\mathrm{cov},j} \to \infty$ tends pointedly to $\mathbb{R}^4$. Along the explicit diagonal sequence, the cover systole diverges, physical mesh edges shrink, and the Gaussian gap and logarithmic action vanish.

The same link kinematics admits gauge-invariant Euclidean states whose simple Wilson-loop expectations are respectively one, zero, or any prescribed constant between them. The magnetic configuration therefore selects no confinement law. Likewise, entanglement does not define a spectral subspace and imposes no energy floor, while exact agreement with one gapped Gaussian range leaves open a gapless orthogonal channel. Theorem 2.19 isolates the sufficient missing datum: a norm-nondegenerate energy comparison defined on the entire physical excitation form domain.

Nothing in this diagram identifies the finite Gaussian Hilbert space with a physical gauge-invariant compact-link Hilbert space. In particular, the vanishing upper bound for the Gaussian gap is a theorem about the explicitly defined quadratic model, not a conclusion about the spectrum of a nonperturbative Yang–Mills Hamiltonian. This non-identification is part of the mathematical result, not a deferred step.

**Provenance.** The compact-link core was extracted from the canonical consolidated workbench source `cusp_compact_link_gaussian.tex`. The fibrewise cover spectrum and pointed-limit results, and the native continuum and finite Coulomb comparisons, were synchronized from the proved workbench sources `cusp_spectral_geometry.tex` and `cusp_gauge_operators.tex` and restated here with their required definitions and proofs. The smooth integral-flux construction, closed background-gauge domains, and charged negative mode were synchronized from `cusp_su2_magnetic.tex`; they are restated here rather than invoked as an external lemma. The operator-theoretic countermodels and coercive comparison in §2.9 were cross-checked against the internal exact audit module `agent_logs/nested_sector_gap_theorem_20260903.tex` and are reproved here. The workbench sources themselves are not modified.

# References

[1] Levent Alpöge. The $(3, 4, \infty)$ modular family of 2-tori, completed at its three special points, is a complex structure on $s^6$. Unpublished online manuscript, 108 pp.; produced with Claude and also circulated under the title "A compact complex threefold fibred by tori over the projective line, and the six-sphere", 2026.

[2] Martin Lüscher. Construction of a selfadjoint, strictly positive transfer matrix for euclidean lattice gauge theories. *Communications in Mathematical Physics*, 54(3):283–292, 1977.

[3] Pietro Menotti and Andrea Pelissetto. General proof of osterwalder–schrader positivity for the wilson action. *Communications in Mathematical Physics*, 113:369–373, 1987.

[4] Konrad Osterwalder and Robert Schrader. Axioms for euclidean green's functions. *Communications in Mathematical Physics*, 31(2):83–112, 1973.

[5] Konrad Osterwalder and Robert Schrader. Axioms for euclidean green's functions ii. *Communications in Mathematical Physics*, 42(3):281–305, 1975.

[6] Kenneth G. Wilson. Confinement of quarks. *Physical Review D*, 10(8):2445–2459, 1974.